Statistics in Medicine

WILEY

RESEARCH ARTICLE

A generalized gamma kernel-machine promotion-time cure model

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Received: **Revised:** **Accepted:**

Keywords: cure model | generalized gamma | kernel machine | promotion time | survival analysis

ABSTRACT

Cure models make the long-term event-free fraction an explicit quantity, but promotion-time cure models usually take the mean number of latent causes to be log-linear in the covariates, which cannot represent nonlinear, threshold, or interaction effects and can bias both the covariate effects and the estimated cured fraction. We propose a generalized gamma kernel-machine promotion-time cure model (GG-KM) that places the log-intensity of the latent-cause process in a reproducing-kernel Hilbert space, retains a flexible parametric generalized gamma latency distribution, and allows the latent number of causes to be Poisson or negative binomial, thereby accommodating both equidispersion and overdispersion. Estimation uses penalized expectation-maximization for the Poisson model and expectation/conditional maximization either for the negative-binomial model, followed by bounded observed-likelihood refinement and explicit convergence diagnostics. We evaluate the framework through simulation experiments with linear, quadratic, and periodic covariate effects and a four-covariate mechanism design, and through applications to melanoma and breast-cancer cohorts, using event-stratified training and test splits, training-only tuning, and inverse-probability-of-censoring weighted integrated Brier scores together with calibration and cure-fraction accuracy. No single predictor is uniformly best: a correctly specified simpler predictor performs best when the covariate structure matches its form, whereas GG-KM is competitive across mechanisms, is the most accurate predictor for the radial four-covariate mechanism, and improves on the log-linear and additive-spline predictors for the interaction mechanism, although the correctly aligned quadratic model remains best in that setting. In the breast-cancer cohort, the spline models attained the lowest mean integrated Brier score, whereas the mixture-cure model attained the highest mean cure AUC. These findings support GG-KM as a flexible and robust alternative when the functional form of the latent-cause mechanism is unknown, and the melanoma and breast-cancer applications illustrate its use for reproducible out-of-sample prediction.

1 | Introduction

Standard survival models assign vanishing probability to remaining event free as follow-up grows arbitrarily long. That limit is inappropriate when a clinically meaningful subgroup will never experience the event of interest, a situation indicated by a sustained plateau in the survival curve under mature follow-up. Cure models describe this setting by separating the long-term event-free probability from the timing of events among individuals who remain susceptible, which makes the cured proportion an interpretable quantity in oncology and other long-term-survivor studies[1,2].

Accommodating this plateau requires a model that allows a positive limiting survival probability, and three main constructions are available in the literature for this purpose, differing in the mechanism by which they generate the cure mass. The first, the mixture cure model, introduces a latent binary indicator that

separates long-term event-free individuals from those who remain susceptible, so that the cured proportion and the survival of the susceptible subpopulation are specified as separate components, typically a regression model for incidence together with a proper distribution for the latency [3,1,4,5]. The second, the defective-distribution model, does not represent the cured fraction through a separate component; instead, the cured fraction arises intrinsically from the parameter space of a single parametric survival distribution. For most parameter values the distribution is proper and the survival function decays to zero, as in conventional survival models; over a distinct region of the same parameter space the distribution becomes improper and the survival function instead converges to a strictly positive limit, which is the cured fraction. A single family then governs both the failure times of the susceptible individuals and the size of the cured group, with no separate incidence model and with covariates entering through its parameters [6,7,8]. The third, based on latent competing risks, generates the cure mass from an unobserved count of competing causes, so that the cured fraction is The probability that this count is zero [9,10,2,11]. This mechanism admits a two-stage description. An initiation stage sets the unobserved number N of latent causes carried by an individual, and a maturation stage assigns each cause an activation time; the earliest activation produces the event, so $N = 0$ describes an individual who never fails and identifies the cure mass [12]. Encoding the count law through its probability-generating function links the latency distribution to both the population survival and the cure fraction, and it is this representation on which the model developed here is built.

In the promotion-time construction the mean number of latent causes is almost always taken to be log-linear in the covariates [10,13]. This specification is convenient and keeps the cure fraction in closed form, but it forces every covariate to act monotonically on the latent-cause intensity, so threshold effects, interactions, and periodic or otherwise nonlinear associations cannot be represented. Because the intensity governs the cure fraction, a misspecified incidence model biases both the estimated covariate effects and the cured proportion [14]. The activation-time component carries another issue: fixing the latency to a restrictive family can be equally misspecified, and in a cure model this error propagates to the covariate effects and to the cure fraction. A flexible parametric family mitigates this while preserving interpretability and a tractable likelihood, and the generalized gamma is a natural choice, since it nests the exponential, Weibull, gamma, and log-normal distributions and spans a wide range of hazard shapes [15,16,17]. Together, these concerns motivate treating both components flexibly, through a flexible latent-intensity predictor for the incidence and a sufficiently general parametric latency for the activation times.

Flexible predictors have been proposed in cure modelling: smoothers, support-vector machines, neural networks, and deep architectures have all been used to relax the usual linear specification. In the promotion-time setting, Pal and Aselisewine [18] introduced a support-vector formulation for the incidence component, and Xie and Yu [19] used a neural network for an unstructured promotion-time component, while deep promotion-time models and structured nonlinear cure models further broaden the available high-dimensional architectures [20,21]. These methods make the covariate effect on the latent-cause intensity nonlinear, but they differ substantially in how they treat that intensity, the latency distribution, and the count structure. To our knowledge, no existing promotion-time approach combines all three within a single likelihood-based estimation and prediction framework: a flexible nonlinear predictor for the latent-cause intensity, which shapes both the cured fraction and the timing of events; a transparent and sufficiently general parametric latency; and an explicit count-family treatment of overdispersion.

To address this gap, we develop a generalized gamma kernel-machine promotion-time cure model (GG-KM) that places the log-intensity of the latent-cause process in a reproducing-kernel Hilbert space (RKHS) [22], while retaining a flexible parametric generalized gamma latency distribution [15] and allowing either Poisson or negative-binomial latent counts. This formulation separates two sources of modelling flexibility: the kernel predictor captures potentially complex nonlinear and interaction effects of covariates on the latent-cause mechanism, whereas the generalized gamma distribution provides a flexible parametric representation of the activation-time process.

The remainder of this article is organized as follows. Section 2 presents the proposed generalized gamma kernel-machine promotion-time cure model, covering the general promotion-time construction, the generalized gamma latency distribution, the kernel-based latent-cause predictor, and the penalized EM and ECME estimation procedures with their observed-likelihood refinement and convergence assessment. Section 3 describes the simulation study and evaluates the predictive performance of the proposed model against log-linear, additive-spline, and quadratic alternatives under different covariate structures and latent-count distributions. Section 4 applies the model to the melanoma data and reports its predictive behaviour. Section 6 discusses the main findings, limitations, and directions for future research.

2 | Methods

This section first recalls, as background, the general promotion-time cure construction that the model builds on, and then develops the proposed GG-KM model: a generalized gamma latency, a regularized reproducing-kernel Hilbert space predictor for the latent-cause intensity, and Poisson or negative-binomial latent counts. The remaining subsections give the penalized estimation algorithms and the prediction and evaluation protocol. Observed likelihoods and conditional objective functions are stated in the main text, while lengthy derivations and analytical gradients are deferred to the Supporting Information.

2.1 | General promotion-time cure construction

Let T_i denote the event time and C_i the right-censoring time. The observed time is the random variable $Y_i = \min(T_i, C_i)$, whose realized value is written t_i, and $\delta_i = I(T_i \le C_i)$ is the event indicator, with $\boldsymbol{x}_i$ a covariate vector, for $i = 1, \dots, n$. The observed data are therefore $\mathcal{D}_{\text{obs}} = \{(t_i, \delta_i, \boldsymbol{x}_i)\}_{i=1}^{n}$. Censoring is assumed conditionally independent, $T_i \perp C_i \mid \boldsymbol{x}_i$. The latent variable N_i counts the potential causes of the event, and $N_i = 0$ identifies a cured individual.

Conditional on the covariates, $N_i \mid \boldsymbol{x}_i$ has probability mass function $\pi_k(\boldsymbol{x}_i) = P(N_i = k \mid \boldsymbol{x}_i)$, $k = 0, 1, 2, \dots$, and probability-generating function $A_i(u) = \mathbb{E}\left(u^{N_i} \mid \boldsymbol{x}_i\right) = \sum_{k \ge 0} \pi_k(\boldsymbol{x}_i) u^k$. Given $N_i = k \ge 1$, the activation times $Z_{i1}, \dots, Z_{ik}$ are independent and

identically distributed with a proper distribution function F_0, survival function $S_0 = 1 - F_0$, and density f_0; conditional on $\boldsymbol{x}_i$ they are independent of N_i, and their distribution does not depend on N_i. In the formulation considered here the covariates act on the latent-count distribution but not on the latency F_0. The event time is the earliest activation, $T_i = \min_{1 \le j \le N_i} Z_{ij}$ when $N_i \ge 1$, whereas $T_i = \infty$ when $N_i = 0$.

Conditional on $N_i = k$, the event exceeds t only if all k activation times do, so $P(T_i > t \mid N_i = k, \boldsymbol{x}_i) = S_0(t)^k$. Averaging over the latent-count distribution gives the population survival function and, as $t \to \infty$ with $S_0(t) \to 0$, the cure fraction,

$$\begin{aligned} S_{\text{pop}}(t \mid \boldsymbol{x}_i) &= P(T_i > t \mid \boldsymbol{x}_i) = \sum_{k \ge 0} \pi_k(\boldsymbol{x}_i) S_0(t)^k = A_i\{S_0(t)\}, \\ p_c(\boldsymbol{x}_i) &= \lim_{t \to \infty} S_{\text{pop}}(t \mid \boldsymbol{x}_i) = A_i(0) = P(N_i = 0 \mid \boldsymbol{x}_i). \end{aligned} \tag{1}$$

The population survival is the generating function evaluated at the latency survival, and the cure fraction is its value at zero, equal to the probability of no latent cause. Differentiating with respect to time, and using $S_0' = -f_0$, gives the population density and hazard,

$$\begin{aligned} f_{\text{pop}}(t \mid \boldsymbol{x}_i) &= f_0(t) A_i'\{S_0(t)\}, \\ h_{\text{pop}}(t \mid \boldsymbol{x}_i) &= \frac{f_{\text{pop}}(t \mid \boldsymbol{x}_i)}{S_{\text{pop}}(t \mid \boldsymbol{x}_i)}, \end{aligned} \tag{2}$$

where A_i' denotes differentiation with respect to its argument. When $p_c(\boldsymbol{x}_i) > 0$, f_{pop} is an improper subdensity that integrates to $1 - p_c(\boldsymbol{x}_i)$ rather than to one. The construction satisfies the required boundary behaviour: $S_{\text{pop}}(0 \mid \boldsymbol{x}) = 1$, $S_{\text{pop}}(t \mid \boldsymbol{x})$ is non-increasing in t, and $\lim_{t \to \infty} S_{\text{pop}}(t \mid \boldsymbol{x}) = p_c(\boldsymbol{x})$.

Equations (1) and (2) give the standard promotion-time construction for a generic activation-time distribution. The proposed GG-KM model, developed in the following subsection, specializes the latency F_0 to the Stacy generalized gamma distribution and models the latent-count mean $\theta_i = \theta(\boldsymbol{x}_i)$ through a reproducing-kernel Hilbert space predictor, with the count taken to be Poisson or negative binomial.

2.2 | The proposed GG-KM model

The activation time of each latent cause follows the Stacy generalized gamma distribution with scale $a > 0$ and shape parameters $d > 0$ and $p > 0$.[15] Its density, distribution function, and survival function are

$$\begin{aligned} f_{GG}(t; a, d, p) &= \frac{p t^{d-1} \exp\{-(t/a)^p\}}{a^d \Gamma(d/p)}, \\ F_{GG}(t; a, d, p) &= P\{d/p, (t/a)^p\}, \\ S_{GG}(t; a, d, p) &= Q\{d/p, (t/a)^p\}, \end{aligned} \tag{3}$$

where P and Q are the regularized lower and upper incomplete gamma functions. This parameterization contains Weibull models when $d = p$, gamma models when $p = 1$, and the exponential model when $d = p = 1$. The lognormal distribution is not a special case or limit of the Stacy parameterization at fixed (a, d, p); obtaining it requires a separate reparameterized limit and is therefore not claimed here. The implementation verifies numerically that the density integrates to one and that the derivative of the distribution function equals the density, so F_{GG}, f_{GG}, and S_{GG} remain mutually consistent. Figure 1 illustrates how the verified special cases and a general parameter setting produce distinct density and hazard shapes.

The GG-KM model relaxes the usual log-linear specification of the latent-cause intensity by representing $\eta(\boldsymbol{x}) = \log\{\theta(\boldsymbol{x})\}$ through a reproducing kernel Hilbert space (RKHS). Here, $\theta(\boldsymbol{x}) > 0$ denotes the conditional mean of the latent number of causes,

$$\theta(\boldsymbol{x}) = \mathbb{E}(N \mid \boldsymbol{x}), \tag{4}$$

under both the Poisson and negative-binomial parameterizations considered in this work. Modelling $\eta(\boldsymbol{x})$ in an RKHS allows the association between the covariates and the latent-cause mechanism to be nonlinear without requiring a prespecified finite-dimensional functional form.

A semiparametric extension of the generalized representer theorem[23] allows an RKHS component to be combined with an unpenalized finite-dimensional parametric component. Let $\mathcal{H}_0 = \text{span}\{\psi_1, \dots, \psi_M\}$, where $\psi_1, \dots, \psi_M : \mathcal{X} \to \mathbb{R}$, and define

$$\Psi_{i\ell} = \psi_\ell(\boldsymbol{x}_i), \qquad i \in [n], \quad \ell \in [M].$$

Assume that the resulting $n \times M$ matrix $\boldsymbol{\Psi}$ has full column rank M, so that the coefficients of the finite-dimensional component are identifiable on the observed sample.

Consider predictors of the form

$$\widetilde{f} = f + h, \qquad f \in \mathcal{H}_K, \qquad h \in \mathcal{H}_0,$$

and suppose that estimation is based on minimizing a criterion of the form

$$\mathcal{C}\left(\widetilde{f}(\boldsymbol{x}_1), \dots, \widetilde{f}(\boldsymbol{x}_n)\right) + G\left(\|f\|_{\mathcal{H}_K}\right),$$

where the data are regarded as fixed arguments of $\mathcal{C}$ and $G : [0, \infty) \to \mathbb{R}$ is strictly increasing. Under these conditions, the generalized representer theorem implies that the RKHS component of any minimizer can be represented using only kernel sections evaluated at the training observations. Consequently, a minimizer admits the representation

$$\widetilde{f}^\star(\boldsymbol{x}) = \sum_{j=1}^{n} \alpha_j K(\boldsymbol{x}, \boldsymbol{x}_j) + \sum_{\ell=1}^{M} \beta_\ell \psi_\ell(\boldsymbol{x}). \tag{5}$$

Thus, regularization is applied only to the RKHS component, whereas the finite-dimensional component remains unpenalized.

For the GG-KM model, estimation is formulated as the minimization of a penalized negative log-likelihood over the latent-cause predictor η. Let $\ell(\eta; \mathcal{D}_{\text{obs}})$ denote the corresponding observed-data log-likelihood. The corresponding penalized estimation problem is

$$\min_{\eta} \left\{ -\ell(\eta; \mathcal{D}_{\text{obs}}) + \frac{\lambda}{2} \|f\|^2_{\mathcal{H}_K} \right\}, \qquad \lambda > 0, \tag{6}$$

where the penalization is applied only to the RKHS component of η.

To place this optimization problem within the semiparametric representer-theorem framework, the unpenalized finite-dimensional space is chosen to contain only an intercept. Specifically, setting $M = 1$, $\psi_1(\boldsymbol{x}) = 1$, so that $\mathcal{H}_0 = \text{span}\{1\}$.

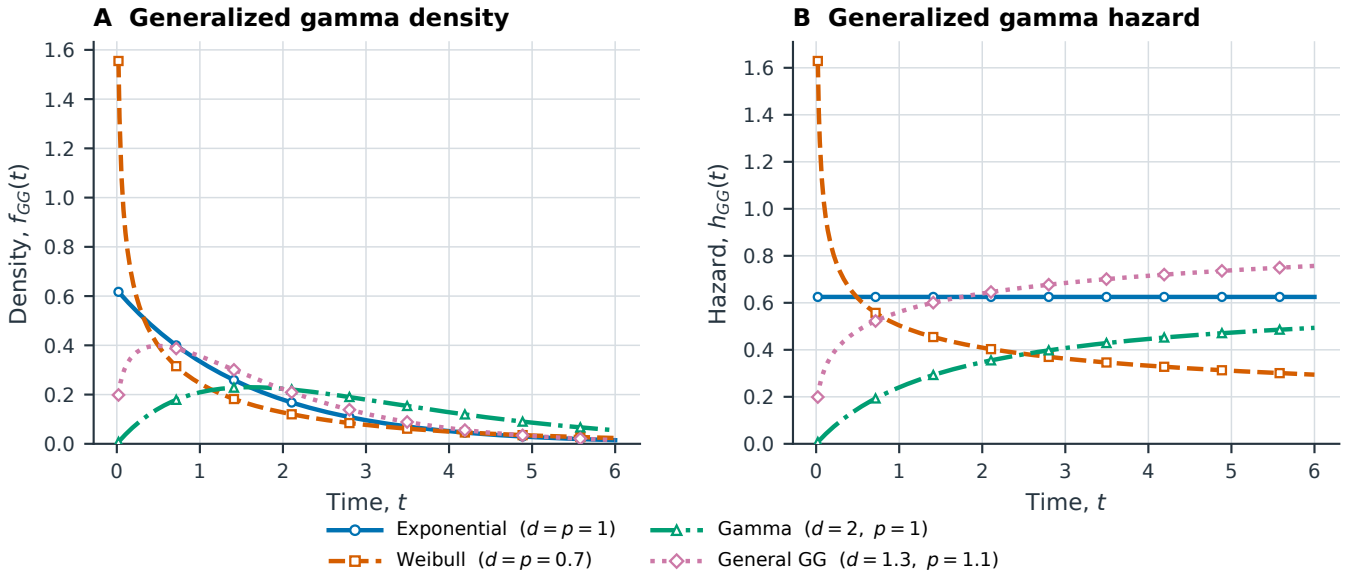


FIGURE 1 | Generalized gamma latency flexibility under the Stacy parameterization. Panel A shows the density $f_{GG}(t; a, d, p)$ and Panel B the corresponding hazard $h_{GG}(t; a, d, p) = f_{GG}(t; a, d, p)/S_{GG}(t; a, d, p)$ for the exponential case $(a, d, p) = (1.6, 1, 1)$, Weibull case $(1.6, 0.7, 0.7)$, gamma case $(1.6, 2, 1)$, and the general generalized gamma setting $(1.6, 1.3, 1.1)$ used in the simulation study. Line type and markers distinguish the four cases in addition to colour.

Consequently, the class of predictors over which Equation (6) is minimized is

$$\eta(\boldsymbol{x}) = \beta_0 + f(\boldsymbol{x}), \qquad \beta_0 \in \mathbb{R}, \qquad f \in \mathcal{H}_K. \tag{7}$$

Thus, the intercept constitutes the unpenalized finite-dimensional component, whereas f contains the possibly nonlinear covariate effect represented in the RKHS. In this case, the matrix $\boldsymbol{\Psi}$ reduces to the n-vector of ones and has rank one for every nonempty sample.

Substituting Equation (7) into the likelihood, the estimation problem can be written explicitly as

$$(\widehat{\beta}_0, \widehat{f}) = \arg \min_{\beta_0 \in \mathbb{R},\, f \in \mathcal{H}_K} \left\{ -\ell(\beta_0, f; \mathcal{D}_{\text{obs}}) + \frac{\lambda}{2} \|f\|^2_{\mathcal{H}_K} \right\}. \tag{8}$$

Here, $\ell(\beta_0, f; \mathcal{D}_{\text{obs}})$ denotes the same observed-data log-likelihood evaluated through the predictor $\eta(\boldsymbol{x}) = \beta_0 + f(\boldsymbol{x})$. The regularization term in Equation (8) is of the form required by the representer theorem. Indeed, for

$$G(\|f\|_{\mathcal{H}_K}) = \frac{\lambda}{2} \|f\|^2_{\mathcal{H}_K}, \qquad \|f\|_{\mathcal{H}_K} \geq 0,$$

G is strictly increasing when $\lambda > 0$. Therefore, the semiparametric representer theorem applies to the minimization problem in Equation (8). It follows that the RKHS component of any minimizer can be expressed as $\widehat{f}(\boldsymbol{x}) = \sum_{j=1}^{n} \widehat{\alpha}_j K(\boldsymbol{x}, \boldsymbol{x}_j)$. Hence, although Equation (8) is initially posed as an optimization problem over the infinite-dimensional space $\mathcal{H}_K$, its solution can be sought within the finite-dimensional family

$$\widehat{\eta}(\boldsymbol{x}) = \widehat{\beta}_0 + \sum_{j=1}^{n} \widehat{\alpha}_j K(\boldsymbol{x}, \boldsymbol{x}_j). \tag{9}$$

Let $\boldsymbol{K}$ denote the $n \times n$ training Gram matrix, with entries $K_{ij} = K(\boldsymbol{x}_i, \boldsymbol{x}_j)$. For $f(\cdot) = \sum_{j=1}^{n} \alpha_j K(\cdot, \boldsymbol{x}_j)$, the squared RKHS norm satisfies $\|f\|^2_{\mathcal{H}_K} = \boldsymbol{\alpha}^\top \boldsymbol{K} \boldsymbol{\alpha}$. Accordingly, the penalized estimation problem reduces to the finite-dimensional optimization

$$\min_{\beta_0 \in \mathbb{R},\, \boldsymbol{\alpha} \in \mathbb{R}^n} \left\{ -\ell(\beta_0, \boldsymbol{\alpha}; \mathcal{D}_{\text{obs}}) + \frac{\lambda}{2} \boldsymbol{\alpha}^\top \boldsymbol{K} \boldsymbol{\alpha} \right\}. \tag{10}$$

Thus, increasing λ imposes stronger regularization on the nonparametric RKHS component of the predictor, while the intercept β_0 remains unpenalized.

Finally, the GG-KM predictor is linked to the conditional mean of the latent number of causes through

$$\theta(\boldsymbol{x}) = \exp\{\eta(\boldsymbol{x})\}. \tag{11}$$

Therefore, $\eta(\boldsymbol{x})$ is unrestricted on $\mathbb{R}$, whereas the exponential link guarantees $\theta(\boldsymbol{x}) > 0$ without requiring an explicit positivity constraint.

The count law converts the intensity into a population survival function and a cure fraction through its generating function. Writing $\theta_i = \theta(\boldsymbol{x}_i)$, Poisson counts have generating function

$$A_i(u) = \exp\{\theta_i(u - 1)\},$$

so that substituting $u = S_{GG}(t)$ in Equation (1) for the population survival and $u = 0$ for the cure fraction gives

$$\begin{aligned} S_{\text{P}}(t \mid \boldsymbol{x}) &= \exp\{-\theta(\boldsymbol{x}) F_{GG}(t)\}, \\ p_{\text{c,P}}(\boldsymbol{x}) &= \exp\{-\theta(\boldsymbol{x})\}, \end{aligned} \tag{12}$$

with population density and hazard

$$\begin{aligned} f_{\text{pop}}(t \mid \boldsymbol{x}) &= \theta(\boldsymbol{x}) f_{GG}(t) \exp\{-\theta(\boldsymbol{x}) F_{GG}(t)\}, \\ h_{\text{pop}}(t \mid \boldsymbol{x}) &= \theta(\boldsymbol{x}) f_{GG}(t), \end{aligned}$$

so the Poisson population hazard is the latency density scaled by the intensity, and $\text{Var}(N_i \mid \boldsymbol{x}_i) = \theta_i$. Negative-binomial counts

with dispersion $\phi > 0$ have generating function

$$A_i(u) = \{1 + \phi\theta_i(1-u)\}^{-1/\phi},$$

with mean θ_i and variance $\theta_i(1 + \phi\theta_i)$, so ϕ controls overdispersion relative to the Poisson case; the same substitutions give

$$\begin{aligned} S_{\mathrm{NB}}(t \mid \boldsymbol{x}) &= \{1 + \phi\theta(\boldsymbol{x})F_{GG}(t)\}^{-1/\phi}, \\ p_{c,\mathrm{NB}}(\boldsymbol{x}) &= \{1 + \phi\theta(\boldsymbol{x})\}^{-1/\phi}, \end{aligned} \tag{13}$$

with population density and hazard

$$\begin{aligned} f_{\mathrm{pop}}(t \mid \boldsymbol{x}) &= \theta(\boldsymbol{x})f_{GG}(t)\{1 + \phi\theta(\boldsymbol{x})F_{GG}(t)\}^{-1/\phi-1}, \\ h_{\mathrm{pop}}(t \mid \boldsymbol{x}) &= \frac{\theta(\boldsymbol{x})f_{GG}(t)}{1 + \phi\theta(\boldsymbol{x})F_{GG}(t)}. \end{aligned}$$

The negative-binomial law reduces to the Poisson law as $\phi \to 0$, so that $S_{\mathrm{NB}} \to S_{\mathrm{P}}$ and $p_{c,\mathrm{NB}} \to p_{c,\mathrm{P}}$, giving a direct check on the two count models. The cure fraction is strictly positive at every finite $\theta(\boldsymbol{x})$ and approaches one only as $\theta(\boldsymbol{x}) \to 0$.

2.3 | Estimation Procedures

Estimation of the GG-KM model is based on the EM algorithm, with regularization incorporated through a penalized EM procedure. The Poisson specification is estimated using penalized EM, whereas the negative-binomial specification is estimated using the Expectation/Conditional Maximization Either (ECME) algorithm[24]. The latter is particularly useful for the negative-binomial model because the additional dispersion parameter ϕ leads to a more complicated complete-data maximization step.

Let $\mathcal{D}_{\mathrm{comp}} = \{(t_i, \delta_i, N_i, \boldsymbol{x}_i)\}_{i=1}^n$ denote the corresponding complete data obtained by augmenting the observations with the latent number of causes N_i. The parameter vectors for the Poisson and negative-binomial models are respectively $\boldsymbol{\Theta}_{\mathrm{P}} = (\beta_0, \boldsymbol{\alpha}, a, d, p)$ and $\boldsymbol{\Theta}_{\mathrm{NB}} = (\beta_0, \boldsymbol{\alpha}, a, d, p, \phi)$. For both specifications, $\eta_i = \eta(\boldsymbol{x}_i) = \beta_0 + \boldsymbol{K}_i^{\top}\boldsymbol{\alpha}$, $\theta_i = \exp(\eta_i)$, where $\boldsymbol{K}_i = \big(K(\boldsymbol{x}_i, \boldsymbol{x}_1), \ldots, K(\boldsymbol{x}_i, \boldsymbol{x}_n)\big)^{\top}$.

The target criterion is the observed penalized log-likelihood, obtained by combining the count distribution, the generalized gamma latency distribution in Equation (3), the kernel predictor in Equation (9), and the RKHS penalty. For the Poisson model,

$$\ell_{\mathrm{P}}^{\mathrm{pen}}(\boldsymbol{\Theta}_{\mathrm{P}}) = \sum_{i=1}^n [\delta_i\{\eta_i + \log f_{GG}(t_i)\} - \theta_i F_{GG}(t_i)] - \frac{\lambda}{2}\boldsymbol{\alpha}^{\top}\boldsymbol{K}\boldsymbol{\alpha}. \tag{14}$$

For the negative-binomial model,

$$\begin{aligned} \ell_{\mathrm{NB}}^{\mathrm{pen}}(\boldsymbol{\Theta}_{\mathrm{NB}}) = \sum_{i=1}^n \Big[\delta_i\{\eta_i + \log f_{GG}(t_i)\} \\ - \Big(\frac{1}{\phi} + \delta_i\Big)\log\{1 + \phi\theta_i F_{GG}(t_i)\}\Big] - \frac{\lambda}{2}\boldsymbol{\alpha}^{\top}\boldsymbol{K}\boldsymbol{\alpha}, \end{aligned} \tag{15}$$

with $\phi > 0$. Here and throughout this subsection, $f_{GG}(t_i)$, $F_{GG}(t_i)$, and $S_{GG}(t_i)$ are evaluated at the current generalized gamma parameters (a, d, p).

Direct optimization of these observed-data criteria is complicated by the latent counts N_i. The EM construction instead uses the conditional expectation of the complete-data log-likelihood. Given the current parameter estimate $\boldsymbol{\Theta}^{(r)}$, define

$$Q_\lambda\left(\boldsymbol{\Theta} \mid \boldsymbol{\Theta}^{(r)}\right) = \mathbb{E}_{\boldsymbol{\Theta}^{(r)}}\left[\ell_c\left(\boldsymbol{\Theta}; \mathcal{D}_{\mathrm{comp}}\right) \Big| \mathcal{D}_{\mathrm{obs}}\right] - \frac{\lambda}{2}\boldsymbol{\alpha}^{\top}\boldsymbol{K}\boldsymbol{\alpha}. \tag{16}$$

The E-step therefore replaces functions of the unobserved N_i by their conditional expectations under the current parameter values.

For the Poisson specification, the complete-data log-likelihood, up to terms that do not depend on the model parameters, is

$$\begin{aligned} \ell_{c,\mathrm{P}} = \sum_{i=1}^n \{N_i\eta_i - \theta_i\} \\ + \sum_{i=1}^n \{\delta_i \log f_{GG}(t_i) + (N_i - \delta_i)\log S_{GG}(t_i)\}. \end{aligned} \tag{17}$$

The omitted terms, such as those involving the factorial of N_i, are constant with respect to $\boldsymbol{\Theta}_{\mathrm{P}}$ when the complete data are held fixed.

At iteration r, the conditional expectation of the latent count for the Poisson model is

$$\widehat{N}_{i,\mathrm{P}}^{(r)} = \mathbb{E}\left[N_i \mid t_i, \delta_i, \boldsymbol{x}_i; \boldsymbol{\Theta}_{\mathrm{P}}^{(r)}\right] = \delta_i + \theta_i^{(r)} S_{GG}^{(r)}(t_i). \tag{18}$$

For the negative-binomial specification, the corresponding conditional expectation is

$$\widehat{N}_{i,\mathrm{NB}}^{(r)} = \delta_i + \frac{(1 + \phi^{(r)}\delta_i)\theta_i^{(r)} S_{GG}^{(r)}(t_i)}{1 + \phi^{(r)}\theta_i^{(r)} F_{GG}^{(r)}(t_i)}. \tag{19}$$

Once the conditional counts have been computed, the expected complete-data objective separates naturally into a predictor block and a generalized gamma latency block. For notational simplicity, the iteration index on $\widehat{N}_i$ is suppressed in the following expressions.

For the Poisson model, the terms depending on $(\beta_0, \boldsymbol{\alpha})$ are

$$Q_{\eta,\mathrm{P}} = \sum_{i=1}^n \{\widehat{N}_i\eta_i - \exp(\eta_i)\} - \frac{\lambda}{2}\boldsymbol{\alpha}^{\top}\boldsymbol{K}\boldsymbol{\alpha}. \tag{20}$$

For the negative-binomial model, conditional on the current value of ϕ, the corresponding predictor block is

$$Q_{\eta,\mathrm{NB}} = \sum_{i=1}^n \left[\widehat{N}_i\eta_i - \left(\widehat{N}_i + \frac{1}{\phi}\right)\log\{1 + \phi\exp(\eta_i)\}\right] - \frac{\lambda}{2}\boldsymbol{\alpha}^{\top}\boldsymbol{K}\boldsymbol{\alpha}, \tag{21}$$

up to terms that do not depend on $(\beta_0, \boldsymbol{\alpha})$.

For fixed conditional counts, both count models share the same generalized gamma latency block,

$$Q_{GG} = \sum_{i=1}^n \{\delta_i \log f_{GG}(t_i) + (\widehat{N}_i - \delta_i)\log S_{GG}(t_i)\}. \tag{22}$$

Consequently, $(\beta_0, \boldsymbol{\alpha})$ appears only in the corresponding predictor block, whereas (a, d, p) appears only in Equation (22) once $\widehat{N}_i$ is held fixed. This separation permits the predictor and latency parameters to be updated in distinct maximization blocks without altering the underlying penalized EM objective.

For the Poisson specification, these conditional maximizations constitute the penalized EM update. The negative-binomial specification contains the additional dispersion parameter ϕ. Its

complete-data count contribution contains terms involving ϕ and gamma functions of the latent count, which makes direct maximization of the expected complete-data criterion with respect to ϕ considerably less convenient. The ECME algorithm avoids this additional expectation by updating ϕ through the observed-data log-likelihood rather than through the corresponding expected complete-data block.

Specifically, after updating $(\beta_0, \boldsymbol{\alpha})$ and (a, d, p) conditionally on the current dispersion, the ECME update is

$$\phi^{(r+1)} = \arg\max_{\phi>0} \ell_{\mathrm{NB}}^{\mathrm{pen}}\left(\beta_0^{(r+1)}, \boldsymbol{\alpha}^{(r+1)}, a^{(r+1)}, d^{(r+1)}, p^{(r+1)}, \phi\right). \tag{23}$$

Because the RKHS penalty does not depend on ϕ, this update is equivalent to maximizing the unpenalized observed-data log-likelihood with respect to ϕ while all remaining parameters are held fixed.

The Supporting Information provides the complete-data derivations, the conditional expectations in Equations (18)–(19), and the analytical gradients used to optimize the predictor and latency blocks.

Algorithms 1 and 2 summarize the penalized EM and ECME procedures for the Poisson and negative-binomial GG-KM models, respectively.

The simulations and application developed throughout this work use different families of kernels to represent distinct forms of similarity between covariate vectors. Let $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X} \subseteq \mathbb{R}^r$. The kernel families considered are

$$\begin{aligned}
K_{\mathrm{lin}}(\boldsymbol{x}, \boldsymbol{x}') &= \boldsymbol{x}^{\top}\boldsymbol{x}', \\
K_{\mathrm{poly}}(\boldsymbol{x}, \boldsymbol{x}') &= (\gamma\, \boldsymbol{x}^{\top}\boldsymbol{x}' + c_0)^q, \\
K_{\mathrm{rbf}}(\boldsymbol{x}, \boldsymbol{x}') &= \exp\{-\gamma\|\boldsymbol{x} - \boldsymbol{x}'\|_2^2\}, \\
K_{\mathrm{exp}}(\boldsymbol{x}, \boldsymbol{x}') &= \exp\{-\gamma\|\boldsymbol{x} - \boldsymbol{x}'\|_2\}, \\
K_{\mathrm{lap}}(\boldsymbol{x}, \boldsymbol{x}') &= \exp\{-\gamma\|\boldsymbol{x} - \boldsymbol{x}'\|_1\}, \\
K_{\mathrm{cau}}(\boldsymbol{x}, \boldsymbol{x}') &= \{1 + \gamma\|\boldsymbol{x} - \boldsymbol{x}'\|_2^2\}^{-1}, \\
K_{\mathrm{wav}}(\boldsymbol{x}, \boldsymbol{x}') &= \prod_{j=1}^{r}\left[\cos\left(1.75\,\frac{x_j - x_j'}{A}\right)\exp\left\{-\frac{1}{2}\left(\frac{x_j - x_j'}{A}\right)^2\right\}\right].
\end{aligned} \tag{24}$$

Here, $\gamma > 0$ is a kernel scale parameter, q is the polynomial degree, c_0 is the polynomial offset, and $A > 0$ controls the scale of the wavelet kernel. The RBF kernel corresponds to the Gaussian kernel and depends on the squared Euclidean distance, whereas the exponential and Laplacian kernels use the Euclidean and L_1 distances, respectively. Kernel-specific hyperparameters are selected within the corresponding tuning procedure.

The GG-KM model is implemented in the open-source Python package `scikit-cure`, available at GitHub.

2.4 | Prediction, tuning, and evaluation

Prediction for an unseen covariate vector uses the training expansion $\eta(\boldsymbol{x}_{\mathrm{new}}) = \widehat{\beta}_0 + K(\boldsymbol{x}_{\mathrm{new}}, \boldsymbol{X}_{\mathrm{train}})\widehat{\boldsymbol{\alpha}}$. Equations (12) and (13) then produce the individual cure fraction and population survival curve. Covariate preprocessing is estimated from the training data and applied unchanged to validation and test data.

Predictive evaluation uses common event-stratified outer splits with 70% of observations for training and 30% for testing. Kernel parameters, the penalty, model variants, and all preprocessing choices are selected inside the outer training set by a shared inner tuning protocol and comparable search budget. The same outer test observations and evaluation grid are used for every paired method comparison.

Let $\widehat{G}$ be the Kaplan–Meier estimate of the censoring survival distribution fitted from the outer training data, with censoring treated as the event. The weighting convention follows the prediction-error construction of Graf and colleagues.[25] For a test set of size m, the inverse-probability-of-censoring weighted Brier score is

$$\widehat{\mathrm{BS}}(t) = \frac{1}{m}\sum_{i=1}^{m}\left\{\frac{I(t_i \le t, \delta_i = 1)}{\widehat{G}(t_i-)}\widehat{S}_i(t)^2 + \frac{I(t_i > t)}{\widehat{G}(t)}[1 - \widehat{S}_i(t)]^2\right\}. \tag{25}$$

The integration horizon is selected from the outer training data and restricted to times with reliable censoring support. The integrated Brier score is the time average obtained by trapezoidal integration of Equation (25) over the frozen grid and division by the horizon length, without renormalizing by the realized sum of weights.

Discrimination with respect to long-term cure status is additionally assessed using an imputation-based area under the receiver operating characteristic curve for cure models, denoted by $\mathrm{AUC}_{\mathrm{cure}}$[20]. Let

$$\widehat{\pi}_i = \lim_{t\to\infty}\widehat{S}_i(t) \tag{26}$$

denote the estimated cure probability for subject i. For the Poisson GG-KM model, $\widehat{\pi}_i = \exp\{-\exp(\widehat{\eta}_i)\}$, whereas for the negative-binomial GG-KM model, $\widehat{\pi}_i = \left(1 + \widehat{\phi}\exp(\widehat{\eta}_i)\right)^{-1/\widehat{\phi}}$. Because cure status is not directly observed for censored individuals, $\widehat{\pi}_i$ and $1-\widehat{\pi}_i$ are used as imputation weights for the cured and susceptible states, respectively. For a cut-off $c \in [0, 1]$, the estimated true-positive and false-positive rates are

$$\widehat{\mathrm{TPR}}(c) = \frac{\sum_{i=1}^{m} I(\widehat{\pi}_i \le c)(1 - \widehat{\pi}_i)}{\sum_{i=1}^{m}(1 - \widehat{\pi}_i)}, \tag{27}$$

$$\widehat{\mathrm{FPR}}(c) = \frac{\sum_{i=1}^{m} I(\widehat{\pi}_i \le c)\widehat{\pi}_i}{\sum_{i=1}^{m}\widehat{\pi}_i}. \tag{28}$$

Here, lower values of $\widehat{\pi}_i$ correspond to stronger evidence that an individual is susceptible, so susceptibility is treated as the positive class. The $\mathrm{AUC}_{\mathrm{cure}}$ is obtained by trapezoidal integration of the receiver operating characteristic curve formed by Equations (27) and (28), that is $\{\widehat{\mathrm{FPR}}(c), \widehat{\mathrm{TPR}}(c)\}$ over $c \in [0, 1]$. Larger values indicate better discrimination between individuals with higher and lower probabilities of belonging to the susceptible subpopulation. Evaluation summaries retain paired score differences, cure-fraction error in simulations, $\mathrm{AUC}_{\mathrm{cure}}$, calibration, convergence, failures, and runtime. A common generalized gamma latency is used throughout the reported simulation so that the generating latency matches the fitted latency structure and predictor-shape contrasts are not confounded with unmodelled latency heterogeneity. Covariate-dependent latency was not evaluated in the

Algorithm 1 Penalized EM algorithm for the Poisson GG-KM model

Require: Training data $(\boldsymbol{X}, \boldsymbol{t}, \boldsymbol{\delta})$, kernel configuration, $\lambda > 0$, tolerance $\varepsilon > 0$, and maximum number of iterations $k_{\max}$

Construct $\boldsymbol{K}$ and initialize $(\beta_0^{(0)}, \boldsymbol{\alpha}^{(0)}, \boldsymbol{\xi}^{(0)})$, with $\boldsymbol{\xi} = (\log a, \log d, \log p)^\top$; evaluate $\ell_{\mathrm{P}}^{\mathrm{pen},(0)}$.

for $k = 0, \ldots, k_{\max} - 1$ do

  Set $\eta_i^{(k)} = \beta_0^{(k)} + \boldsymbol{K}_i^\top \boldsymbol{\alpha}^{(k)}$ and $(a^{(k)}, d^{(k)}, p^{(k)}) = \exp\{\boldsymbol{\xi}^{(k)}\}$.

  E-step: for $i = 1, \ldots, n$, compute

$$\widehat{N}_i^{(k+1)} = \delta_i + \exp\{\eta_i^{(k)}\} S_{GG}\big(t_i; a^{(k)}, d^{(k)}, p^{(k)}\big).$$

  M-step: with $\widehat{N}_i^{(k+1)}$ fixed, update

$$(\beta_0^{(k+1)}, \boldsymbol{\alpha}^{(k+1)}) = \arg\max_{\beta_0, \boldsymbol{\alpha}} \left\{ \sum_{i=1}^{n} \left[\widehat{N}_i^{(k+1)} \eta_i - e^{\eta_i} \right] - \frac{\lambda}{2} \boldsymbol{\alpha}^\top \boldsymbol{K} \boldsymbol{\alpha} \right\}.$$

  Update the latency parameters on the log scale:

$$\boldsymbol{\xi}^{(k+1)} = \arg\max_{\boldsymbol{\xi}} \sum_{i=1}^{n} \left[\delta_i \log f_{GG}\big(t_i; e^{\xi_a}, e^{\xi_d}, e^{\xi_p}\big) + (\widehat{N}_i^{(k+1)} - \delta_i) \log S_{GG}\big(t_i; e^{\xi_a}, e^{\xi_d}, e^{\xi_p}\big) \right].$$

  Set $(a^{(k+1)}, d^{(k+1)}, p^{(k+1)}) = \exp\{\boldsymbol{\xi}^{(k+1)}\}$ and evaluate $\ell_{\mathrm{P}}^{\mathrm{pen},(k+1)}$.

  if $\left| \ell_{\mathrm{P}}^{\mathrm{pen},(k+1)} - \ell_{\mathrm{P}}^{\mathrm{pen},(k)} \right| \le \varepsilon$ then

    break

  end if

end for

return $(\widehat{\beta}_0, \widehat{\boldsymbol{\alpha}}, \widehat{a}, \widehat{d}, \widehat{p})$.

Algorithm 2 Penalized ECME algorithm for the negative-binomial GG-KM model

Require: Training data $(\boldsymbol{X}, \boldsymbol{t}, \boldsymbol{\delta})$, kernel configuration, $\lambda > 0$, tolerance $\varepsilon > 0$, and maximum number of iterations $k_{\max}$

Construct $\boldsymbol{K}$ and initialize $(\beta_0^{(0)}, \boldsymbol{\alpha}^{(0)}, \boldsymbol{\xi}^{(0)}, \zeta^{(0)})$, with $\boldsymbol{\xi} = (\log a, \log d, \log p)^\top$ and $\zeta = \log \phi$; evaluate $\ell_{\mathrm{NB}}^{\mathrm{pen},(0)}$.

for $k = 0, \ldots, k_{\max} - 1$ do

  Set $\eta_i^{(k)} = \beta_0^{(k)} + \boldsymbol{K}_i^\top \boldsymbol{\alpha}^{(k)}$, $(a^{(k)}, d^{(k)}, p^{(k)}) = \exp\{\boldsymbol{\xi}^{(k)}\}$, and $\phi^{(k)} = e^{\zeta^{(k)}}$.

  E-step: for $i = 1, \ldots, n$, compute

$$\widehat{N}_i^{(k+1)} = \delta_i + \frac{(1 + \phi^{(k)} \delta_i) e^{\eta_i^{(k)}} S_{GG}(t_i; a^{(k)}, d^{(k)}, p^{(k)})}{1 + \phi^{(k)} e^{\eta_i^{(k)}} F_{GG}(t_i; a^{(k)}, d^{(k)}, p^{(k)})}.$$

  M-step: with $\widehat{N}_i^{(k+1)}$ and $\phi^{(k)}$ fixed, update

$$(\beta_0^{(k+1)}, \boldsymbol{\alpha}^{(k+1)}) = \arg\max_{\beta_0, \boldsymbol{\alpha}} \left\{ \sum_{i=1}^{n} \left[\widehat{N}_i^{(k+1)} \eta_i - \left(\widehat{N}_i^{(k+1)} + \frac{1}{\phi^{(k)}} \right) \log\{1 + \phi^{(k)} e^{\eta_i}\} \right] - \frac{\lambda}{2} \boldsymbol{\alpha}^\top \boldsymbol{K} \boldsymbol{\alpha} \right\}.$$

  Update the latency parameters on the log scale:

$$\boldsymbol{\xi}^{(k+1)} = \arg\max_{\boldsymbol{\xi}} \sum_{i=1}^{n} \left[\delta_i \log f_{GG}(t_i; e^{\xi_a}, e^{\xi_d}, e^{\xi_p}) + (\widehat{N}_i^{(k+1)} - \delta_i) \log S_{GG}(t_i; e^{\xi_a}, e^{\xi_d}, e^{\xi_p}) \right].$$

  ECME-step: update the dispersion on the log scale by maximizing the observed penalized log-likelihood:

$$\zeta^{(k+1)} = \arg\max_{\zeta \in \mathbb{R}} \ell_{\mathrm{NB}}^{\mathrm{pen}} \left(\beta_0^{(k+1)}, \boldsymbol{\alpha}^{(k+1)}, e^{\boldsymbol{\xi}^{(k+1)}}, e^{\zeta} \right).$$

  Set $(a^{(k+1)}, d^{(k+1)}, p^{(k+1)}) = \exp\{\boldsymbol{\xi}^{(k+1)}\}$ and $\phi^{(k+1)} = e^{\zeta^{(k+1)}}$; evaluate $\ell_{\mathrm{NB}}^{\mathrm{pen},(k+1)}$.

  if $\left| \ell_{\mathrm{NB}}^{\mathrm{pen},(k+1)} - \ell_{\mathrm{NB}}^{\mathrm{pen},(k)} \right| \le \varepsilon$ then

    break

  end if

end for

return $(\widehat{\beta}_0, \widehat{\boldsymbol{\alpha}}, \widehat{a}, \widehat{d}, \widehat{p}, \widehat{\phi})$.

reported scenarios and remains a future misspecification stress test.

3 | Simulation study

The simulation study was designed to evaluate the predictive performance of the proposed GG-KM model under different cure mechanisms, latent-count distributions, and sample sizes. Specifically, we investigated three questions. First, we examined whether a simpler predictor performs best when its functional form is correctly specified. Second, we evaluated whether GG-KM provides robust predictions when the true cure-probability function is not adequately represented by the simpler models. Third, we assessed the extent to which predictive performance depends on the latent-count distribution and its degree of overdispersion. The generalized gamma latency distribution was correctly specified and held fixed across all scenarios. Therefore, differences in predictive performance primarily reflect the ability of each predictor to recover the covariate-dependent cure mechanism.

Each data-generating scenario was defined by one cure-probability surface, one latent-count distribution, and one sample size. We considered three cure-probability surfaces, linear, quadratic, and periodic. We also considered three latent-count distributions, Poisson, negative binomial with dispersion $\phi = 0.25$, and negative binomial with dispersion $\phi = 1.00$. The two negative-binomial settings represent modest and stronger overdispersion, respectively, whereas the Poisson distribution represents the equidispersed limiting case. The sample sizes were $n = 300$ and $n = 600$, resulting in 18 data-generating scenarios.

In every scenario, two covariates were generated independently as $x_1, x_2 \sim \text{Uniform}(-1, 1)$. The bounded uniform domain deliberately differs from the historical source because it gives all three geometries a common domain for level-set and recovery comparisons. The source used standard-normal covariates and a covariate-dependent Weibull latency; retaining those choices would confound cure-predictor shape with unmodelled latency heterogeneity.

The true cure probability was generated from one of three fixed surfaces. Specifically, with $\text{expit}(v) = \{1+\exp(-v)\}^{-1}$, the designs were

$$\begin{aligned} p_c^{\text{lin}}(\boldsymbol{x}) &= \text{expit}(0.15 - 1.00x_1 + 0.75x_2), \\ p_c^{\text{quad}}(\boldsymbol{x}) &= \text{expit}(0.60 - 1.45x_1^2 - 1.05x_2^2 + 0.55x_1x_2), \\ p_c^{\text{per}}(\boldsymbol{x}) &= \text{expit}\{0.10 + 1.15\sin(\pi x_1) + 0.80\cos(\pi x_2)\}. \end{aligned} \tag{29}$$

For Poisson generation, the latent-count mean was $\theta(\boldsymbol{x}) = -\log p_c(\boldsymbol{x})$; for negative-binomial generation, it was $\theta(\boldsymbol{x}) = \{p_c(\boldsymbol{x})^{-\phi} - 1\}/\phi$. This mapping preserves the same cure-probability surface across count distributions, so comparisons do not confound dispersion with a different cure target.

Figure 2 shows the three surfaces on the common covariate square. The linear surface increases along one tilted gradient and has parallel straight level sets, so the log-linear predictor is correctly specified for it. The quadratic surface has closed elliptical level sets generated by the two squared terms and the interaction, and no additive predictor reproduces that geometry. The periodic surface reverses its gradient twice inside the domain and produces alternating regions of high and low cure probability, a shape an additive spline can follow one coordinate at a time but a degree-two map cannot. The three designs therefore separate correct linear specification, smooth nonadditivity and oscillation while holding the latency distribution fixed. They are not matched on prevalence: the mean of $p_c(\boldsymbol{x})$ over the square is 0.533 for the linear surface, 0.448 for the quadratic surface and 0.520 for the periodic surface, with attainable ranges $[0.168, 0.870]$, $[0.079, 0.646]$ and $[0.136, 0.886]$. Scores are consequently comparable between methods within a scenario and not between scenarios.

All latent-cause activation times followed the generalized gamma distribution with $(a, d, p) = (1.6, 1.3, 1.1)$. Independent censoring times were exponential with mean 3.5, and observed time was the minimum of the event and censoring times. The inverse population distribution was used to generate finite event times directly for susceptible individuals, while individuals in the limiting cure mass were assigned no event. Every replicate retained the true individual cure probability and population survival curve for evaluation.

Algorithm 3 states the implemented inverse-transform generator. Sampling from the improper population distribution avoids simulating a potentially large latent count while preserving its cure mass and finite-event distribution exactly.

The fitted comparison set shared the same generalized gamma latency and comprised GG-KM predictors with linear, radial-basis, Laplacian, exponential, Cauchy, polynomial, and wavelet kernels, each with Poisson and negative-binomial latent counts, together with additive cubic-spline Poisson and negative-binomial models and the classical mixture-cure and promotion-time comparators. Linear-kernel candidates used $\lambda \in \{0.3, 3\}$. Radial-basis, Laplacian, exponential, and Cauchy candidates used $\gamma \in \{0.25, 1\}$ with $\lambda \in \{0.3, 3\}$. Polynomial candidates additionally used degree $q \in \{2, 3\}$ with $c_0 = 1$, while wavelet candidates used $A \in \{0.5, 1, 2\}$, in both cases with $\lambda \in \{0.3, 3\}$. Spline candidates used three or five training-fold quantile knots with $\lambda \in \{0.3, 3\}$. All tuned methods used the same two-fold inner tuning criterion and evaluation-grid size. The promotion-time support-vector method of Pal and Aselisewine was not included because the reproducibility audit found no public executable implementation at the article or arXiv records, and a new reimplementation would not provide a controlled software comparison.[18]

Each generated sample was split once into a common event-stratified 70% training set and 30% test set for all fitted methods. Tuning, feature-map fitting, and preprocessing were confined to the training set, and the IPCW integration horizon was set from that set at the smaller of its 90th observed-time percentile and the limit at which estimated censoring survival remained at least 0.05. Primary test-set estimands were integrated Brier score, cure AUC, and cure-probability mean squared error.

A target-size diagnostic pilot used five planned replicates per pilot scenario and was not pooled with the final experiment; at least four replicates were successful in every pilot cell. The largest observed between-replicate standard deviations were 0.0418 for integrated Brier score and 0.0332 for cure-probability mean squared error. Requiring projected Monte Carlo standard errors no larger than 0.003 and 0.005, respectively, implied at least 195 replicates for integrated Brier score and 45 for cure-probability mean squared error. The more restrictive criterion therefore required 195 replicates, so the final count was fixed at 250 replicates per scenario before final estimation. At $R = 250$, the corresponding

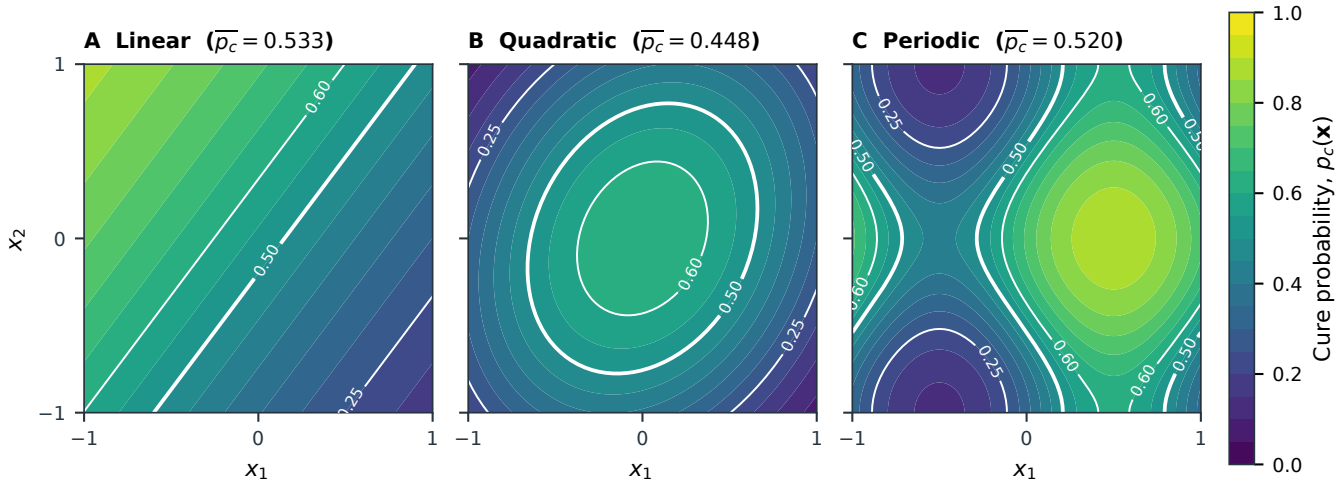


FIGURE 2 | Frozen cure-probability surfaces used in the simulation design. Panels show the linear, quadratic, and periodic functions in Equation (29) over $(x_1, x_2) \in [-1, 1]^2$. All panels use the same $p_c(\boldsymbol{x})$ colour scale; superimposed contour lines provide a colour-independent encoding of the surface geometry. Contours are labelled at $p_c = 0.25, 0.50$ and 0.60, the same levels used in Figure 4, and each panel title reports the mean of $p_c(\boldsymbol{x})$ over the square.

Algorithm 3 Right-censored simulation from a promotion-time population distribution

Require: Scenario, sample size n, generalized gamma parameters (a, d, p), censoring scale, and deterministic replication seed
for $i = 1, \ldots, n$ do
  Draw $x_{i1}, x_{i2} \overset{\text{ind}}{\sim} \text{Uniform}(-1, 1)$ and evaluate the declared $p_c(\boldsymbol{x}_i)$
  Set $\theta_i = -\log p_c(\boldsymbol{x}_i)$ for Poisson counts, or $\theta_i = \{p_c(\boldsymbol{x}_i)^{-\phi} - 1\}/\phi$ for negative-binomial counts
  Draw $U_i \sim \text{Uniform}(0, 1)$ from the population event-time distribution
  if $U_i \geq 1 - p_c(\boldsymbol{x}_i)$ then
    Set $T_i = \infty$ (the draw falls in the cure mass)
  else
    Set $q_i = -\log(1 - U_i)/\theta_i$ for Poisson counts
    Set $q_i = \{(1 - U_i)^{-\phi} - 1\}/(\phi\theta_i)$ for negative-binomial counts
    Set $T_i = F_{GG}^{-1}(q_i; a, d, p)$
  end if
  Draw C_i from the declared independent censoring distribution
  Record the observed time $t_i = \min(T_i, C_i)$, indicator $\delta_i = I(T_i \leq C_i)$, $p_c(\boldsymbol{x}_i)$, and the true population survival grid
end for
Return the complete observed sample, generating truth, scenario identifier, and replication seed

projected Monte Carlo standard errors were 0.00265 for integrated Brier score and 0.00210 for cure-probability mean squared error. For cure AUC, the largest pilot standard deviation was 0.1211, corresponding to a projected Monte Carlo standard error of 0.00766 at $R = 250$. Final summaries report the realized Monte Carlo standard error for every primary metric and paired differences computed within each shared replication.

3.1 | Simulation results

The primary final experiment contained 81,000 method rows from 250 replications in each of 18 scenarios. The largest realized Monte Carlo standard errors were 0.00175 for the IPCW integrated Brier score, 0.00434 for cure-probability RMSE, and 0.00694 for cure AUC. Of the 81,000 fits, 80,842 completed successfully and 158 failed; among the successful fits, 80,832 satisfied the explicit convergence criterion and 10 did not. Method-specific successful-fit totals ranged from 4,467 to 4,498 of 4,500. Mean per-fit runtime ranged from 1.13 to 40.32 seconds across methods, with a maximum of 520.48 seconds.

Table 1 reports the best observed mean IPCW integrated Brier score, cure root mean squared error, and cure AUC in each scenario. For the linear surface, a linear-kernel GG-KM attained the lowest mean IPCW integrated Brier score and cure root mean squared error in all six scenarios. For the quadratic surface, a polynomial-kernel GG-KM attained the lowest cure root mean squared error in all six scenarios and the lowest integrated Brier score in five of six scenarios, with an RBF-kernel GG-KM attaining the lowest integrated Brier score in the remaining case. For the periodic surface, an additive spline model attained the lowest

integrated Brier score and cure root mean squared error in all six scenarios. For cure AUC, a polynomial-kernel GG-KM attained the highest mean in all six linear scenarios and all six periodic scenarios. On the quadratic surface, polynomial-kernel GG-KM models attained the highest mean cure AUC in four of six scenarios, whereas wavelet-kernel GG-KM models attained the highest mean in the remaining two.

Figure 3 shows paired differences between the RBF GG-KM and the count-matched linear-kernel GG-KM, additive-spline, and polynomial-kernel GG-KM comparators across the six generating-law and sample-size cells. Each contrast is defined as the RBF GG-KM value minus the comparator value; hence, negative differences favor the RBF model for IPCW integrated Brier score and cure root mean squared error, whereas positive differences favor it for cure AUC.

For the linear cure surface, the linear-kernel GG-KM consistently outperformed the RBF model in integrated Brier score and cure root mean squared error, while the spline and polynomial comparisons were generally closer. Cure AUC also favored the linear and polynomial kernels, whereas the RBF–spline differences remained comparatively small. Under the quadratic surface, the RBF GG-KM substantially improved on the linear kernel across all three metrics, but its differences from the spline and polynomial models were much smaller; the polynomial kernel generally had lower cure root mean squared error. For the periodic surface, the RBF model improved markedly on the linear and polynomial kernels in integrated Brier score and cure root mean squared error, particularly at $n = 600$, but remained inferior to the spline. Cure AUC showed a different pattern, with the RBF model clearly outperforming the linear kernel and, at $n = 600$, the Poisson polynomial-kernel GG-KM, while its differences from the spline remained comparatively small. Overall, the paired comparisons show that relative performance depends strongly on both the geometry of the cure surface and the evaluation metric.

Figure 4 maps where in the covariate square each predictor loses accuracy. Every panel fills the 41×41 evaluation grid with the pointwise root mean squared error of the predicted cure probability against the generating surface, averaged over the valid Poisson replications at $n = 600$, on a scale shared by all twelve panels. Solid lines repeat the true level sets at $p_c = 0.25$, 0.50 and 0.60, and the dashed line is the mean estimated $p_c = 0.50$ level, which separates an error of magnitude from a displacement of shape. Failed fits reduce the pointwise n_{valid} rather than being removed from the total number of replications. The panel means summarize the gridwise recovery of the cure surface. On the linear surface the linear-kernel GG-KM attains 0.0490, compared with 0.0671 for the additive spline, 0.0694 for the polynomial-kernel GG-KM, and 0.0730 for the RBF GG-KM. On the quadratic surface, the RBF GG-KM attains the lowest mean error, 0.0802, followed closely by the additive spline at 0.0816 and the polynomial-kernel GG-KM at 0.0827, whereas the linear-kernel GG-KM increases to 0.1282. On the periodic surface, the additive spline performs best at 0.0824, followed by the RBF GG-KM at 0.0959, while the linear- and polynomial-kernel GG-KM predictors increase to 0.1504 and 0.1510, respectively. The spatial pattern adds what these scalar summaries omit. A misaligned predictor does not spread its error evenly: the linear-kernel GG-KM shows concentrated error on the quadratic and periodic surfaces, and its estimated $p_c = 0.50$ level is absent on the quadratic surface because the fitted cure surface does not reach that level. The polynomial-kernel GG-KM similarly deteriorates on the periodic surface. In contrast, the RBF GG-KM maintains comparatively stable recovery across the three generating mechanisms relative to the linear and polynomial kernels, with its estimated $p_c = 0.50$ contour closely following the true level wherever it is present. Figure 4 therefore complements the regret summary in Table 2 by showing where prediction errors occur spatially: no single predictor dominates across all three surfaces, with the linear kernel performing best on the linear surface, the RBF kernel on the quadratic surface, and the additive spline on the periodic surface. The complete mean IPCW integrated Brier scores for all eight methods, with their Monte Carlo standard errors, appear in the Supporting Information. They confirm the geometry-specific ordering in Table 1 and retain both fitted count laws for every predictor.

To complement the scenario-specific comparisons, we summarized performance using regret relative to the best-performing method within the retained comparison set. Table 2 restricts this analysis to the Poisson and negative-binomial spline models and RBF GG-KM models across the 18 scenarios. For IPCW integrated Brier score, the Poisson spline had the smallest maximum paired regret, 0.00065 with a 95% Monte Carlo half-width of 0.00056, whereas the negative-binomial spline had the smallest equally weighted mean regret, 0.00012 (half-width 0.00010). For cure root mean squared error, the Poisson spline had the smallest maximum regret, 0.00464 (half-width 0.00210), whereas the negative-binomial spline had the smallest mean regret, 0.00073 (half-width 0.00025). In contrast, the negative-binomial RBF GG-KM was most competitive for cure AUC, attaining the smallest maximum regret, 0.00687 (half-width 0.00464), the smallest mean regret, 0.00144 (half-width 0.00060), and the best descriptive mean rank of 2.06. Thus, the regret summaries favor the spline models for prediction error but show greater competitiveness of the RBF GG-KM for cure discrimination.

3.2 | Higher-dimensional interaction surfaces

The four-covariate experiment is a secondary mechanism study developed after the primary analysis. It probes interaction structures that an additive spline cannot represent, but it is neither independent confirmation nor evidence of a general kernel advantage. Covariates satisfy $x_1, \ldots, x_4 \sim \text{Uniform}(-1, 1)$, and the first target is the *interaction* surface

$$p_c(\boldsymbol{x}) = \text{expit}\big(C_{\text{int}} + 1.60\, x_1 x_2 - 1.30\, x_3 x_4 + 0.70\, x_1 - 0.50\, x_4\big), \tag{30}$$

and the *radial* surface is

$$p_c(\boldsymbol{x}) = \text{expit}\big(C_{\text{rad}} + 3.00 \exp\{-0.80(x_1^2 + x_2^2 + x_3^2 + x_4^2)\}\big). \tag{31}$$

Intercepts C_{int} and C_{rad} were calibrated by bisection on a fixed sample of 200 000 observations (seed 20260901) so that each surface yields a mean cure fraction of 0.45. The latency remained the common generalized gamma with $(a, d, p) = (1.6, 1.3, 1.1)$ and no covariate-dependent scale; censoring was exponential with mean 3.5. The same fitted methods as in the primary experiment were compared under Poisson, negative-binomial ($\phi = 0.25$), and negative-binomial ($\phi = 1.00$) generating laws at $n = 300$ and

TABLE 1 | Best observed mean predictive performance in each simulation scenario among Poisson and negative-binomial models. For IPCW IBS and cure RMSE, smaller values are better, whereas for cure AUC, larger values are better. Parenthesized values are Monte Carlo standard errors. The displayed optimum is a descriptive comparison and is not a claim of superiority when paired uncertainty includes zero. GG, Spl, P, and NB denote generalized-gamma kernel machine, spline, Poisson, and negative binomial, respectively. For GG-KM fits, the selected kernel is shown in parentheses.

True surface	Generating law	n	Lowest mean IPCW IBS	Lowest mean cure RMSE	Highest mean cure AUC
Linear	P	300	GG-NB (linear): 0.1929 (0.0013)	GG-P (linear): 0.0630 (0.0019)	GG-NB (poly): 0.6845 (0.0025)
Linear	NB(0.25)	300	GG-NB (linear): 0.1965 (0.0015)	GG-NB (linear): 0.0602 (0.0017)	GG-NB (poly): 0.6930 (0.0025)
Linear	NB(1.00)	300	GG-NB (linear): 0.1975 (0.0014)	GG-P (linear): 0.0606 (0.0017)	GG-NB (poly): 0.7032 (0.0025)
Linear	P	600	GG-NB (linear): 0.1937 (0.0010)	GG-P (linear): 0.0458 (0.0011)	GG-NB (poly): 0.6875 (0.0020)
Linear	NB(0.25)	600	GG-NB (linear): 0.1944 (0.0010)	GG-P (linear): 0.0448 (0.0011)	GG-NB (poly): 0.6910 (0.0019)
Linear	NB(1.00)	600	GG-NB (linear): 0.1968 (0.0011)	GG-NB (linear): 0.0453 (0.0012)	GG-NB (poly): 0.7045 (0.0018)
Quadratic	P	300	GG-NB (poly): 0.2097 (0.0014)	GG-NB (poly): 0.0990 (0.0018)	GG-NB (poly): 0.6411 (0.0029)
Quadratic	NB(0.25)	300	GG-P (rbf): 0.2128 (0.0014)	GG-NB (poly): 0.0962 (0.0016)	GG-P (poly): 0.6443 (0.0031)
Quadratic	NB(1.00)	300	GG-P (poly): 0.2145 (0.0016)	GG-NB (poly): 0.0933 (0.0018)	GG-P (poly): 0.6586 (0.0031)
Quadratic	P	600	GG-NB (poly): 0.2079 (0.0010)	GG-NB (poly): 0.0720 (0.0013)	GG-P (wavelet): 0.6422 (0.0018)
Quadratic	NB(0.25)	600	GG-NB (poly): 0.2096 (0.0009)	GG-NB (poly): 0.0687 (0.0013)	GG-P (wavelet): 0.6452 (0.0018)
Quadratic	NB(1.00)	600	GG-NB (poly): 0.2128 (0.0010)	GG-NB (poly): 0.0646 (0.0012)	GG-NB (poly): 0.6555 (0.0019)
Periodic	P	300	Spl-NB: 0.1913 (0.0017)	Spl-NB: 0.1091 (0.0017)	GG-NB (poly): 0.7237 (0.0034)
Periodic	NB(0.25)	300	Spl-NB: 0.1906 (0.0015)	Spl-NB: 0.1050 (0.0017)	GG-NB (poly): 0.7216 (0.0033)
Periodic	NB(1.00)	300	Spl-NB: 0.1905 (0.0016)	Spl-NB: 0.1005 (0.0016)	GG-NB (poly): 0.7345 (0.0033)
Periodic	P	600	Spl-NB: 0.1851 (0.0010)	Spl-P: 0.0794 (0.0012)	GG-NB (poly): 0.7301 (0.0023)
Periodic	NB(0.25)	600	Spl-NB: 0.1858 (0.0011)	Spl-NB: 0.0778 (0.0012)	GG-NB (poly): 0.7361 (0.0022)
Periodic	NB(1.00)	600	Spl-P: 0.1878 (0.0012)	Spl-P: 0.0728 (0.0013)	GG-NB (poly): 0.7501 (0.0020)

TABLE 2 | Paired regret and descriptive rank summaries over the 18 primary simulation scenarios for the Poisson and negative-binomial models. The comparison is restricted to spline models and GG-KM models using the RBF kernel. Within each scenario and metric, the reference is the retained method with the best finite mean. For IPCW integrated Brier score and cure RMSE, regret is the method value minus the reference value; for cure AUC, regret is the reference value minus the method value, so that smaller regret is better for all metrics. Regret uses only replication pairs for which both methods are finite. The worst regret identifies its scenario and reports a 95% Monte Carlo half-width; mean regret weights scenarios equally and propagates their Monte Carlo standard errors. Mean rank is based on scenario means and is descriptive.

Metric	Method	Worst paired regret (±95%; scenario)	Mean paired regret (±95%)	Mean rank
IPCW integrated Brier score	Spline Negative binomial	0.00067 ± 0.00056 (Quadratic, P, $n = 600$)	0.00012 ± 0.00010	2.06
IPCW integrated Brier score	Spline Poisson	0.00065 ± 0.00056 (Quadratic, P, $n = 600$)	0.00013 ± 0.00010	2.06
IPCW integrated Brier score	GG-KM Poisson (RBF)	0.00348 ± 0.00119 (Periodic, NB(1.00), $n = 300$)	0.00108 ± 0.00017	3.22
IPCW integrated Brier score	GG-KM Negative binomial (RBF)	0.00359 ± 0.00120 (Periodic, NB(1.00), $n = 300$)	0.00108 ± 0.00017	2.67
Cure RMSE	Spline Negative binomial	0.00464 ± 0.00210 (Quadratic, NB(1.00), $n = 600$)	0.00073 ± 0.00025	1.94
Cure RMSE	Spline Poisson	0.00464 ± 0.00210 (Quadratic, NB(1.00), $n = 600$)	0.00076 ± 0.00025	1.94
Cure RMSE	GG-KM Poisson (RBF)	0.01651 ± 0.00268 (Periodic, NB(1.00), $n = 300$)	0.00631 ± 0.00054	3.17
Cure RMSE	GG-KM Negative binomial (RBF)	0.01627 ± 0.00272 (Periodic, NB(1.00), $n = 300$)	0.00632 ± 0.00053	2.94
Cure AUC	Spline Negative binomial	0.00768 ± 0.00529 (Quadratic, P, $n = 300$)	0.00208 ± 0.00069	2.61
Cure AUC	Spline Poisson	0.00693 ± 0.00538 (Quadratic, P, $n = 300$)	0.00224 ± 0.00070	2.94
Cure AUC	GG-KM Poisson (RBF)	0.00795 ± 0.00464 (Periodic, P, $n = 300$)	0.00157 ± 0.00061	2.39
Cure AUC	GG-KM Negative binomial (RBF)	0.00687 ± 0.00464 (Periodic, P, $n = 300$)	0.00144 ± 0.00060	2.06

$n = 600$, giving 12 scenarios. The final experiment used 250 replicates per scenario, and every declared cell remains in the analysis regardless of its ordering.

Table 3 and Figure 5 report the directed comparisons with paired Monte Carlo uncertainty. Their interpretation is limited to the evaluated interaction and radial mechanisms, and it retains every declared cell, including comparisons that do not favor GG-KM.

The complete per-scenario means with Monte Carlo standard errors, together with the numerical diagnostics, are reported in the Supporting Information.

4 | Melanoma application

The empirical application used the malignant melanoma cohort distributed as the `melanoma` data set in version 1.3-32 of the `boot` R package and documented in the original counting-process source.[26,27] The cohort contains 205 patients operated on in Odense, Denmark between 1962 and 1977, with follow-up recorded through 1977. There were 57 deaths from melanoma, 14 deaths from other causes, and 134 patients alive at the end of follow-up. The analysis treated death from melanoma as the event of interest and censored both administrative survivors and deaths from other causes, so the fitted model is cause specific in its event definition and is not a competing-risks analysis; $p_c(\boldsymbol{x})$ is the probability of no melanoma death from the modeled process, not an all-cause survival probability. The predictor set comprised sex, age at operation, calendar year of operation, the natural logarithm of tumour thickness in millimetres, and ulceration status. Follow-up time was converted from days to years, and no observations or covariate values were missing. Continuous and binary predictors passed through the same training-fold standardization step so that no test-set information entered preprocessing. The versioned analysis copy was confirmed row for row against `boot::melanoma` and audited against its recorded SHA-256 checksum before estimation.

Predictive performance was assessed in 50 event-stratified outer holdouts, each assigning 143 patients to training and 62 to testing. The same split indices were used for every fitted method, and all tuning choices were made within three inner training folds using

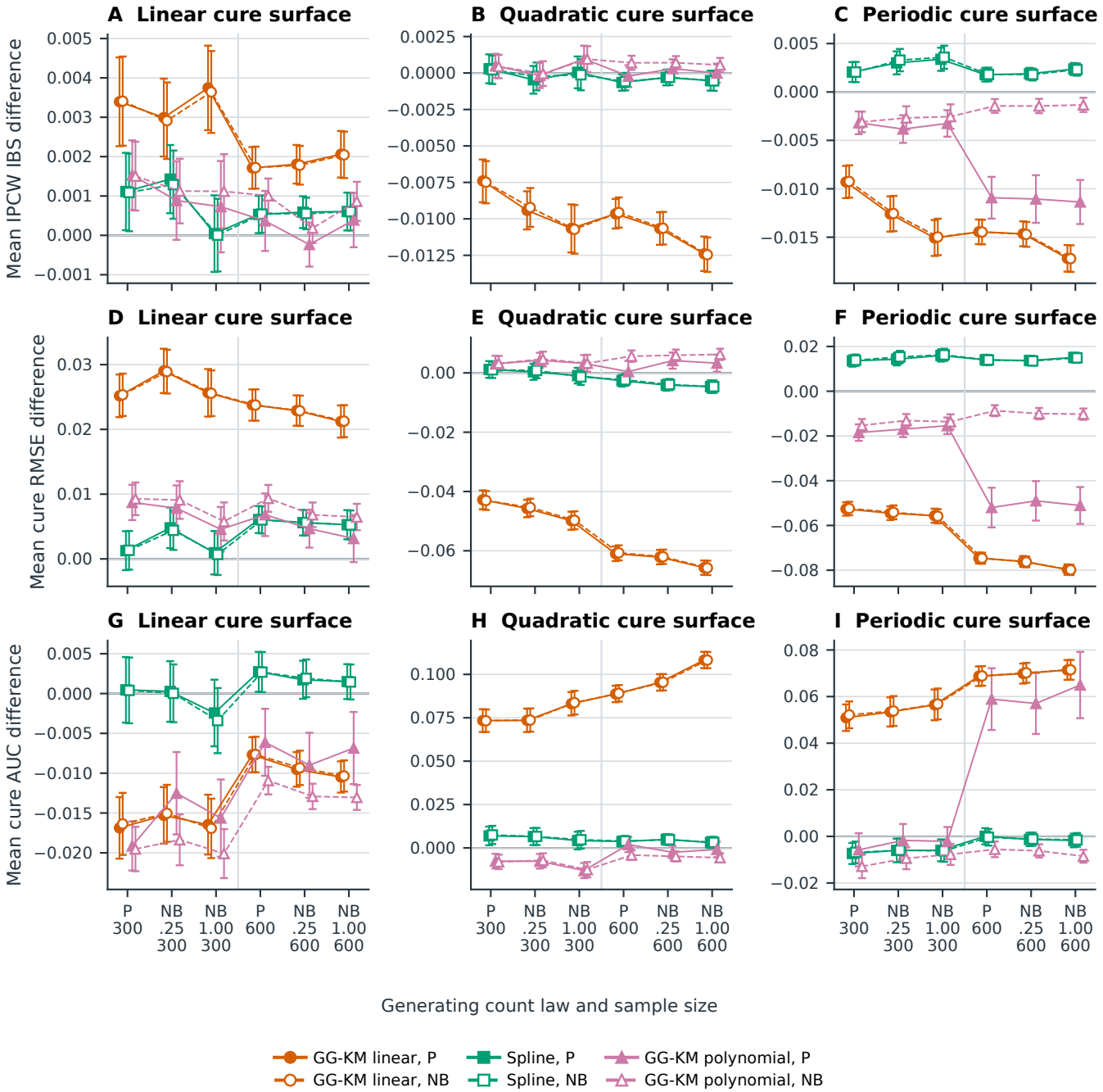


FIGURE 3 | Count-matched paired predictive differences of the RBF GG-KM against the linear-kernel GG-KM, additive-spline, and polynomial-kernel GG-KM comparators. Columns correspond to the linear, quadratic, and periodic cure surfaces, while rows show IPCW integrated Brier score, cure root mean squared error, and cure AUC differences. Each contrast is defined as the RBF GG-KM value minus the corresponding comparator value, with Poisson and negative-binomial fits compared separately. Error bars span ±1.96 paired Monte Carlo standard errors, and a horizontal line marks zero. Negative differences favor the RBF GG-KM for integrated Brier score and cure root mean squared error, whereas positive differences favor it for cure AUC.

the IPCW integrated Brier score and the horizon rule defined in Section 2.4. The comparison comprised GG-KM predictors with linear, radial-basis, Laplacian, exponential, Cauchy, polynomial, and wavelet kernels, each with Poisson and negative-binomial latent-count laws, together with additive cubic-spline Poisson and negative-binomial models and the classical mixture-cure and promotion-time comparators. Kernel and spline hyperparameters were selected from prespecified grids.

Table 4 reports split-level means and variation. The Poisson and negative-binomial GG-KM models with the RBF kernel attained the lowest mean IPCW integrated Brier score, both at 0.1418 with split standard deviation 0.0206, followed closely by the Cauchy kernel under both count laws, with mean IBS 0.1419. The mixture-cure model had the highest mean cure AUC overall, 0.8492, while among the GG-KM models the negative-binomial polynomial kernel had the highest mean cure AUC,

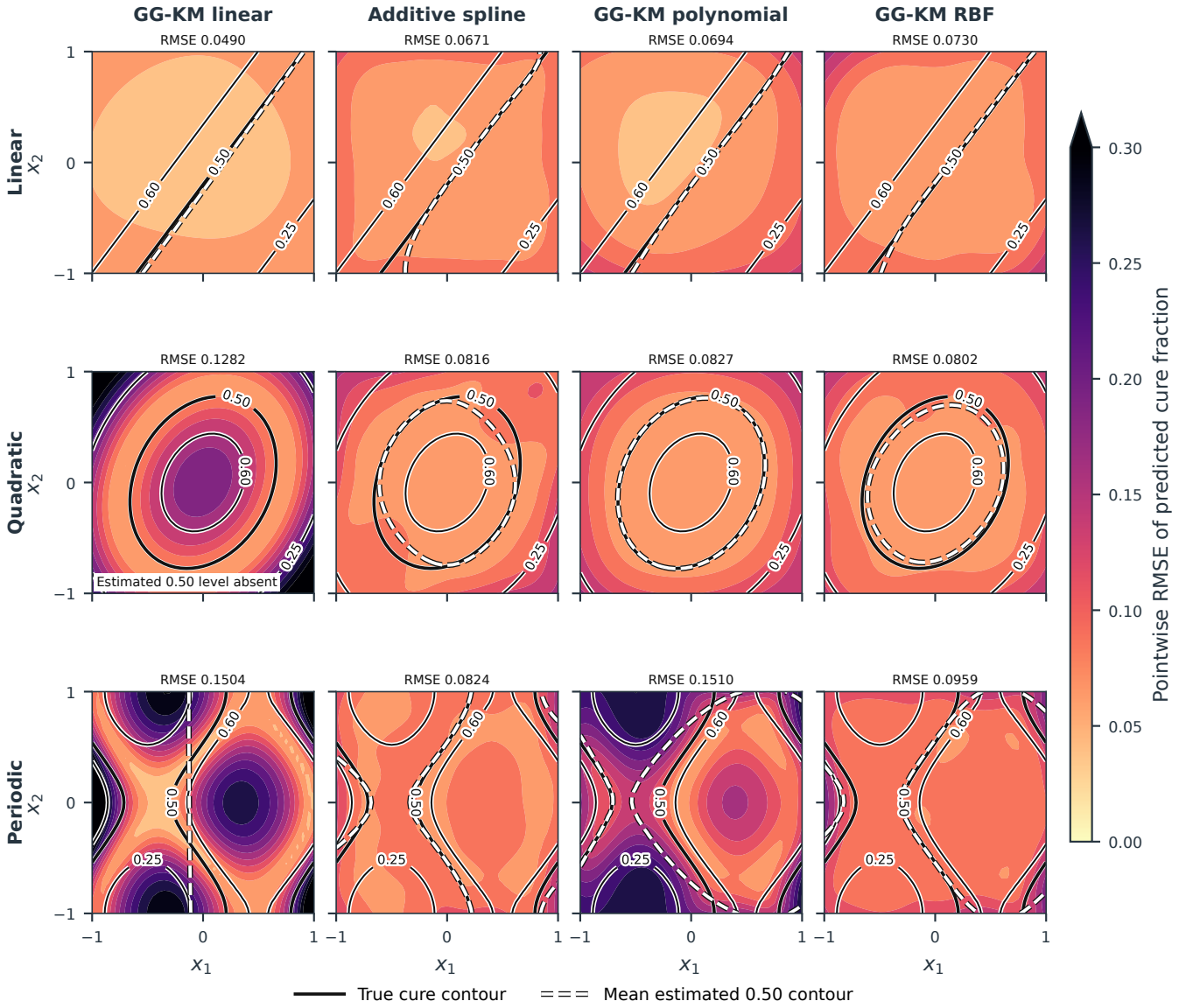


FIGURE 4 | Pointwise accuracy of the three primary cure surfaces at $n = 600$ under Poisson generation. Rows show the linear, quadratic and periodic generating surfaces; columns show the linear-kernel GG-KM, additive-spline, polynomial-kernel GG-KM and radial-basis-function (RBF) GG-KM predictors. The filled field is the root mean squared error of the predicted cure probability at each of the 41×41 grid points, taken across all valid replications, on one scale shared by every panel and truncated above 0.30. Solid labelled lines are the true cure levels 0.25, 0.50 and 0.60; the dashed line is the mean estimated 0.50 level and is annotated as absent when the fitted range does not contain it. The value printed in each panel is the mean of the displayed field. Failed fits remain in n_{total} and reduce the pointwise n_{valid} rather than disappearing.

TABLE 3 | Best observed mean predictive performance in each high-dimensional simulation scenario. For IPCW IBS and cure RMSE, smaller values are better, whereas for cure AUC, larger values are better. Parenthesized values are Monte Carlo standard errors. The comparison includes Poisson and negative-binomial GG-KM models, Poisson and negative-binomial spline models, the mixture-cure model, and the promotion-time model. Int and Rad denote Interaction and Radial surfaces, respectively. GG, Spl, P, and NB denote generalized-gamma kernel machine, spline, Poisson, and negative binomial, respectively. For GG-KM fits, the selected kernel is shown in parentheses.

True surface	Generating law	n	Lowest mean IPCW IBS	Lowest mean cure RMSE	Highest mean cure AUC
Interaction	P	300	GG-NB (polynomial): 0.2045 (0.0017)	GG-NB (polynomial): 0.1250 (0.0017)	GG-NB (polynomial): 0.7052 (0.0024)
Interaction	NB(0.25)	300	GG-NB (polynomial): 0.2075 (0.0015)	GG-NB (polynomial): 0.1254 (0.0017)	GG-NB (polynomial): 0.7179 (0.0024)
Interaction	NB(1.00)	300	GG-NB (polynomial): 0.2093 (0.0017)	GG-NB (polynomial): 0.1179 (0.0018)	GG-NB (polynomial): 0.7289 (0.0023)
Interaction	P	600	GG-NB (polynomial): 0.2018 (0.0011)	GG-NB (polynomial): 0.0961 (0.0014)	GG-NB (polynomial): 0.7043 (0.0017)
Interaction	NB(0.25)	600	GG-NB (polynomial): 0.2027 (0.0011)	GG-NB (polynomial): 0.0912 (0.0013)	GG-NB (polynomial): 0.7091 (0.0017)
Interaction	NB(1.00)	600	GG-NB (polynomial): 0.2027 (0.0011)	GG-NB (polynomial): 0.0869 (0.0012)	GG-P (polynomial): 0.7250 (0.0017)
Radial	P	300	Spl-P: 0.2138 (0.0014)	Spl-P: 0.1168 (0.0017)	GG-NB (polynomial): 0.6521 (0.0023)
Radial	NB(0.25)	300	Spl-P: 0.2167 (0.0014)	Spl-P: 0.1150 (0.0016)	GG-NB (polynomial): 0.6602 (0.0023)
Radial	NB(1.00)	300	Spl-NB: 0.2226 (0.0015)	Spl-NB: 0.1117 (0.0015)	GG-NB (polynomial): 0.6633 (0.0024)
Radial	P	600	Spl-P: 0.2138 (0.0009)	Spl-NB: 0.0926 (0.0011)	GG-NB (polynomial): 0.6426 (0.0019)
Radial	NB(0.25)	600	Spl-P: 0.2136 (0.0009)	Spl-NB: 0.0897 (0.0010)	GG-NB (polynomial): 0.6480 (0.0018)
Radial	NB(1.00)	600	Spl-P: 0.2174 (0.0010)	Spl-P: 0.0881 (0.0010)	GG-NB (polynomial): 0.6492 (0.0018)

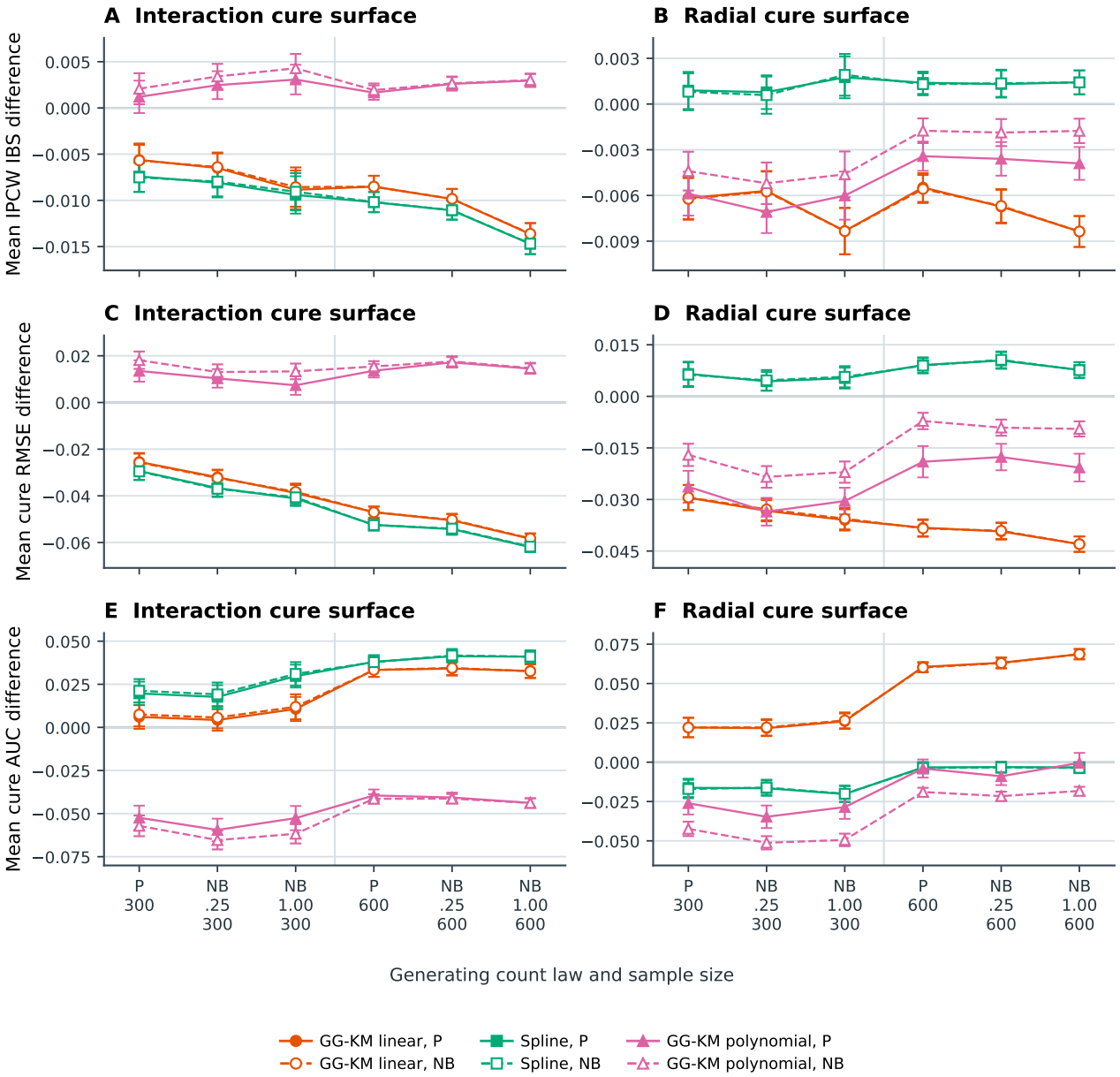


FIGURE 5 | Paired predictive differences of the radial-basis GG-KM predictor against the count-matched GG-KM linear-kernel, spline, and GG-KM polynomial-kernel comparators on the interaction and radial cure surfaces. The top row shows paired mean IPCW integrated Brier score differences and the bottom row paired mean cure root mean squared error differences. Each plotted contrast is defined as the RBF GG-KM value minus the corresponding comparator value, so negative differences favor the RBF GG-KM and positive differences favor the comparator. Solid lines with filled markers denote Poisson fits, whereas dashed lines with open markers denote negative-binomial fits. Each panel displays the six generating-law and sample-size cells, with error bars spanning ±1.96 paired Monte Carlo standard errors.

0.8117. Across the 900 planned split-model fits, 897 completed successfully and 893 satisfied the explicit convergence criteria. Figure 6 places the paired score distributions beside descriptive survival calibration. The comparison of split-averaged predicted survival with the Kaplan–Meier curve is a calibration check on the population survival function; it does not identify individual cure status. Repeated holdouts are dependent because the same patients recur across splits, so empirical quantiles describe variation across training sets rather than independent-sample confidence intervals. Similar mean scores do not establish an approximately linear latent-cause structure, and a smaller split standard deviation alone does not prove greater predictive stability.

Mean predicted cure fractions ranged from 0.572 for the Poisson GG-KM with the wavelet kernel to 0.639 for the Poisson GG-KM with the polynomial kernel. These values are model-based extrapolations because individual cure status is unobserved and follow-up is finite, and the Kaplan–Meier comparison cannot identify a cure fraction. Across-split prediction summaries were retained for every individual to expose variation caused by repeated training samples rather than to create surrogate cure labels. The application consequently evaluates reproducible out-of-sample prediction and numerical behavior while leaving biological interpretation of individual cure probabilities outside its evidential scope.

TABLE 4 | Predictive performance in 50 paired repeated holdouts of the melanoma cohort. Values for the IPCW integrated Brier score (IBS), cure-discrimination area under the curve (AUC_{cure}), and mean predicted cure fraction are means across successful splits, with split standard deviations in parentheses. Convergence is reported as the number of converged fits among successful fits. Runtime is the mean with the maximum in parentheses.

Predictor	Count law	Converged	IPCW IBS	AUC_{cure}	Mean cure prediction	Runtime (s)
Mixture cure	Mixture cure	49/50	0.1430 (0.0185)	0.8492 (0.0491)	0.607 (0.079)	2.28 (12.39)
Promotion time	Poisson	49/49	0.1456 (0.0239)	0.8240 (0.0283)	0.633 (0.047)	2.21 (7.29)
Spline	Poisson	50/50	0.1432 (0.0217)	0.7912 (0.0290)	0.613 (0.049)	42.56 (99.65)
Spline	Negative binomial	50/50	0.1434 (0.0216)	0.7911 (0.0291)	0.614 (0.049)	46.37 (112.34)
GG-KM (linear)	Poisson	50/50	0.1426 (0.0224)	0.8091 (0.0274)	0.627 (0.046)	15.48 (37.21)
GG-KM (linear)	Negative binomial	50/50	0.1426 (0.0224)	0.8091 (0.0274)	0.627 (0.046)	16.29 (38.53)
GG-KM (RBF)	Poisson	50/50	0.1418 (0.0206)	0.6937 (0.0340)	0.586 (0.047)	45.85 (97.33)
GG-KM (RBF)	Negative binomial	50/50	0.1418 (0.0206)	0.6937 (0.0341)	0.586 (0.047)	54.00 (184.25)
GG-KM (Laplacian)	Poisson	48/49	0.1426 (0.0253)	0.6987 (0.0490)	0.578 (0.050)	47.26 (97.01)
GG-KM (Laplacian)	Negative binomial	50/50	0.1438 (0.0257)	0.7021 (0.0526)	0.575 (0.062)	55.12 (140.77)
GG-KM (exponential)	Poisson	49/50	0.1450 (0.0242)	0.6984 (0.0800)	0.580 (0.048)	43.01 (89.43)
GG-KM (exponential)	Negative binomial	50/50	0.1448 (0.0241)	0.6949 (0.0768)	0.580 (0.049)	49.45 (110.53)
GG-KM (Cauchy)	Poisson	50/50	0.1419 (0.0246)	0.6817 (0.0390)	0.585 (0.047)	46.38 (92.49)
GG-KM (Cauchy)	Negative binomial	50/50	0.1419 (0.0246)	0.6823 (0.0390)	0.586 (0.046)	51.95 (173.17)
GG-KM (polynomial)	Poisson	50/50	0.1515 (0.0274)	0.7457 (0.1338)	0.639 (0.063)	113.78 (232.80)
GG-KM (polynomial)	Negative binomial	50/50	0.1477 (0.0280)	0.8117 (0.0261)	0.613 (0.054)	368.69 (806.30)
GG-KM (wavelet)	Poisson	48/49	0.1488 (0.0137)	0.6176 (0.0337)	0.572 (0.061)	64.57 (146.75)
GG-KM (wavelet)	Negative binomial	50/50	0.1482 (0.0131)	0.6193 (0.0320)	0.573 (0.060)	77.38 (219.31)

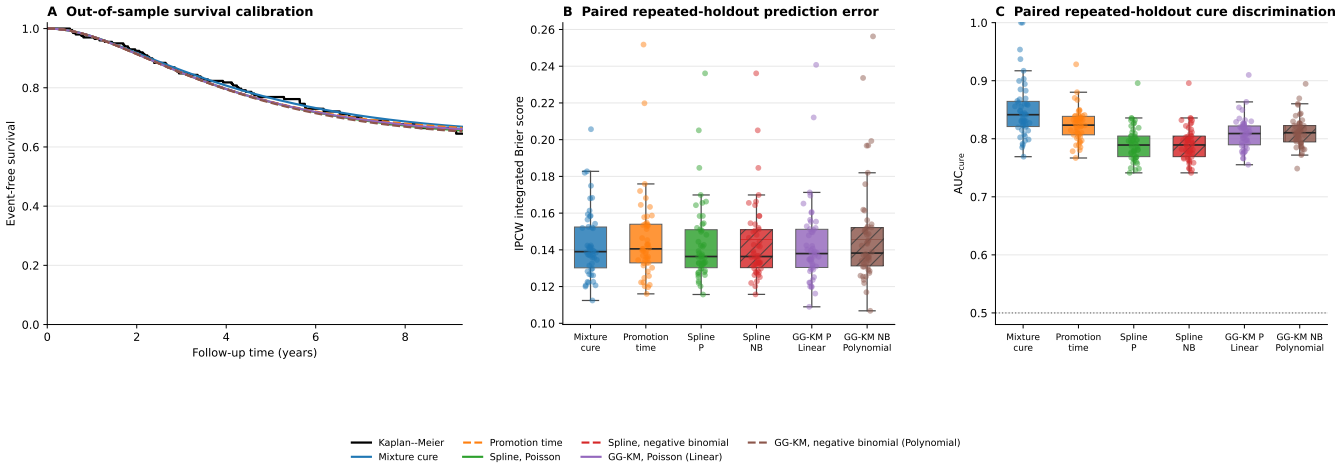


FIGURE 6 | Predictive assessment in 50 paired repeated holdouts of the melanoma cohort. Panel A compares the full-cohort Kaplan–Meier curve with split-averaged out-of-sample survival predictions over the common supported horizon. Panel B shows split-level IPCW integrated Brier scores, with each point representing one outer split and lower values indicating better prediction. Line type, marker or box encoding, and labels distinguish all eight methods without relying on color alone. Panel A shows mean curves without an uncertainty band; the dependence caveat for repeated holdouts is stated in the text.

Figure 7 first indicates a marked similarity between the Poisson and negative-binomial GG-KM specifications. For most kernels, the corresponding boxplots have strongly overlapping distributions for both the IPCW integrated Brier score and AUC_{cure}, and their mean performances are generally close.

For the IPCW integrated Brier score (Panel A), the distributions overlap substantially across kernels, and the differences between the Poisson and negative-binomial versions of the same kernel are generally small. Mean IBS values were relatively close across specifications, with both the Poisson and negative-binomial RBF models attaining the lowest mean value, 0.1418. The Poisson polynomial model had the largest mean IBS, 0.1515, and also showed comparatively large between-split variability. Although several kernels exhibited isolated splits with substantially higher Brier scores, the central parts of the distributions remained similar. Overall, the IBS results do not indicate a clearly dominant kernel.

Panel B shows greater sensitivity to the kernel choice for AUC_{cure}, while still retaining substantial agreement between the two latent-count laws within most kernels. The linear kernel provided consistently strong cure discrimination under both count laws, with mean cure AUCs of approximately 0.809. The negative-binomial polynomial model achieved the highest mean cure AUC among the GG-KM specifications, 0.8117, and showed a relatively concentrated distribution, whereas the corresponding Poisson polynomial model was substantially more variable across splits. The RBF, Laplacian, exponential, and Cauchy kernels generally produced lower cure AUCs, concentrated around 0.68–0.70, while the wavelet kernel showed the weakest discrimination, with mean cure AUCs of approximately 0.62 under both count laws. Thus, the results suggest that kernel specification has a stronger practical influence than the choice between Poisson and

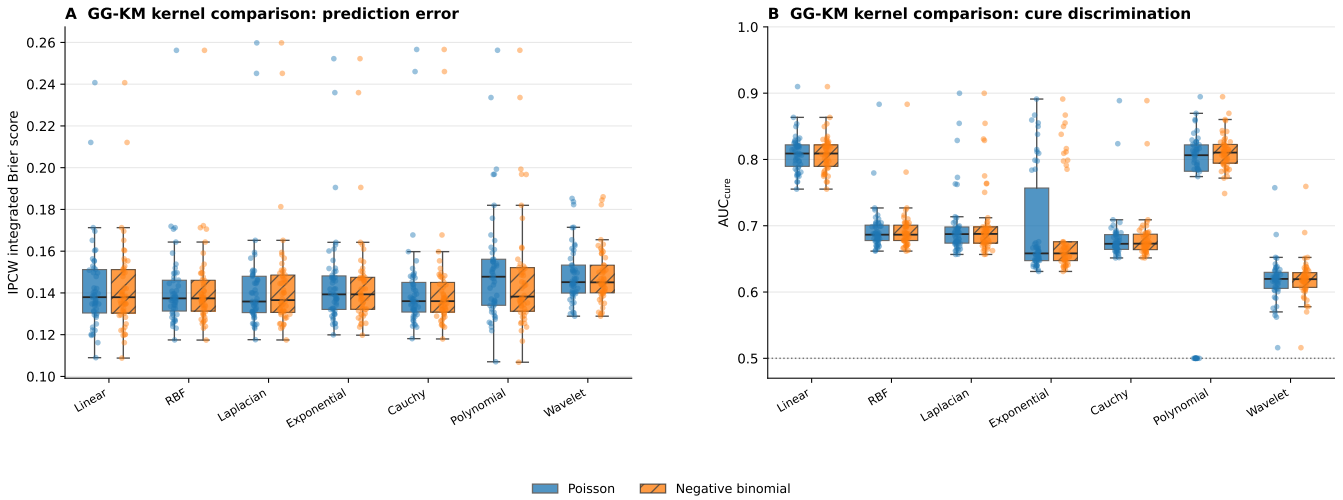


FIGURE 7 | Kernel-specific predictive performance of the GG-KM models in 50 paired repeated holdouts of the melanoma cohort. Panel A shows the split-level IPCW integrated Brier score and Panel B the cure-discrimination area under the curve (AUC_{cure}) across the seven candidate kernels. Within each kernel, Poisson and negative-binomial GG-KM results are shown side by side, with color and hatching distinguishing the two latent-count laws. Each point represents one outer split, and the boxplots summarize the corresponding split-level distributions.

negative-binomial latent-count laws in this application. No single kernel dominated simultaneously in prediction error and cure discrimination: the RBF kernel was most favorable in terms of mean IBS under both count laws, whereas the polynomial and linear kernels provided the strongest cure discrimination.

5 | Breast Cancer Data Application

The second empirical application considered a hospital-based cohort of patients with breast cancer from Paraíba State, Brazil. Follow-up time was represented by the recorded survival time, and death from breast cancer was treated as the event of interest; observations without a recorded breast-cancer death were treated as censored. In contrast with the melanoma application, the breast-cancer data contained a substantially richer set of demographic, socioeconomic, clinical, health-care access, lifestyle, and treatment-related information, allowing the cure component to depend on a broader patient profile.

The predictor set included race or skin-colour classification, educational level, income category, age at first consultation, age group, health macroregion, health-system origin and referral pathway, and the hospital source of the record. Clinical characteristics included primary tumour site, histological type, and clinical stage, with stage grouped into stages I–II, stages III–IV, and a separate category for unavailable staging information. Lifestyle and history variables comprised tobacco use, alcohol use, and family history of cancer. Information on the first treatment received at the hospital was represented through separate indicators for surgery, radiotherapy, chemotherapy, hormone therapy, immunotherapy, no treatment, other treatments, and unavailable treatment information. Health-care timing was represented by the intervals from first consultation to diagnosis, diagnosis to first treatment, and first consultation to first treatment. In addition to their continuous versions, these intervals were represented by indicators distinguishing delays of up to 90 days from delays exceeding 90 days.

Preprocessing was estimated exclusively from each training sample and then applied unchanged to the corresponding validation or test observations. Continuous age and delay variables were transformed using the Yeo–Johnson power transformation [28] and standardized. Explicit missingness indicators were retained for treatment-delay variables, while several categorical predictors included dedicated levels for unavailable information. Because primary tumour site and histological type contained more complex categorical structures, they were represented using cross-validated survival-target encodings based on restricted mean survival time, with smoothing applied within the training data. The resulting representations were carried unchanged to held-out observations, thereby preventing outcome information from the validation or test samples from entering the preprocessing stage. Several prespecified interaction terms were also included to allow clinically relevant associations to vary across patient subgroups. These comprised interactions between clinical stage and first treatment, between stage and hospital source, and between stage and educational level. The continuous consultation-to-diagnosis delay was additionally interacted with stage and with hormone-therapy and immunotherapy indicators. These interaction terms, together with the original transformed continuous variables, survival-target encodings, and categorical indicators, formed the predictor matrix used by all competing models.

Predictive performance was assessed using 50 event-stratified paired outer holdouts, with approximately 70% of the cohort assigned to training and 30% to testing in each split. The same outer split indices were used for all fitted methods. Hyperparameter selection was performed exclusively within the outer training sample using four inner folds and the IPCW integrated Brier score, together with the prediction-horizon rule defined in Section 2.4. The comparison comprised GG-KM models under Poisson and negative-binomial latent-count laws, with linear, radial-basis, Laplacian, exponential, Cauchy, polynomial, and wavelet kernels, as well as spline Poisson and negative-binomial

models and the classical mixture-cure and promotion-time comparators. All kernel and spline hyperparameters were selected from prespecified grids.

Table 5 reports split-level means and variation. The spline models under both Poisson and negative-binomial latent-count laws achieved the lowest mean IPCW integrated Brier score, 0.0877 with split standard deviation 0.0019, followed by the mixture-cure and promotion-time models, both with mean IBS 0.0883. The mixture-cure model attained the highest mean cure AUC overall, 0.7851, followed by the promotion-time model at 0.7775 and the spline models at 0.7707. Among the GG-KM specifications, the negative-binomial linear kernel had the lowest mean IBS, 0.0897, whereas the exponential kernel produced the highest mean cure AUC, 0.7124, under both latent-count laws. Across the 900 planned split-model fits, 873 completed successfully and 871 satisfied the explicit convergence criteria.

Figure 8 summarizes out-of-sample survival calibration, prediction error, and cure discrimination. The split-averaged predicted survival curves closely followed the Kaplan–Meier estimate over the common prediction horizon, with relatively small differences among the fitted methods. The spline Poisson and negative-binomial models showed the lowest IPCW integrated Brier scores, whereas the mixture-cure model achieved the highest cure-discrimination performance. The GG-KM models with the exponential kernel were less competitive in overall prediction error than the spline and classical cure models, but retained substantially better cure discrimination than several alternative kernel specifications.

Figure 9 shows that kernel choice had a marked influence on GG-KM performance. For prediction error, the exponential and Cauchy kernels yielded the lowest mean IBS values among the nonlinear GG-KM specifications, while the wavelet and RBF kernels were less favorable. Cure discrimination varied more substantially across kernels. The exponential kernel produced the highest mean cure AUC among the GG-KM models, 0.7124 under both count laws, whereas the wavelet kernel showed considerably weaker discrimination.

6 | Discussion

The generalized gamma kernel-machine promotion-time cure model extends the promotion-time framework by placing the latent-cause log-intensity in a reproducing-kernel Hilbert space while retaining a flexible generalized gamma latency and both Poisson and negative-binomial latent counts. The primary and secondary simulations and the melanoma and breast-cancer applications evaluate this construction against simpler predictors under a shared protocol. Their interpretation is limited to the evaluated designs. Kernel families were compared within prespecified grids, although robustness to broader kernel and hyperparameter choices remains for future work.

The principal simulation pattern was consistent across the two latent-count laws. On the quadratic surface, the RBF GG-KM lowered the paired IPCW integrated Brier score against the linear-kernel GG-KM by about 0.007 to 0.012 and the paired cure root mean squared error by about 0.04 to 0.07, while its differences from the spline were much smaller and scenario-dependent. On the periodic surface, the RBF GG-KM improved substantially on the linear-kernel GG-KM yet trailed the spline by a small margin, and on the linear surface it paid a small resolved penalty against the correctly specified linear-kernel GG-KM. Within the original three-surface design, the count law was a secondary factor, since Poisson and negative-binomial variants within a predictor family differed by at most 0.02 in integrated Brier score and 0.01 in cure root mean squared error. The melanoma cohort produced overlapping score distributions in which the linear-kernel Poisson model had the lowest mean integrated Brier score and GG-KM was competitive with the smallest split-to-split variation. The breast-cancer cohort showed a complementary pattern: the spline models achieved the lowest mean integrated Brier score, 0.0877, whereas the mixture-cure model attained the highest mean cure AUC, 0.7851. Among the GG-KM specifications, the negative-binomial linear kernel had the lowest mean integrated Brier score, 0.0897, while the exponential kernel attained the highest mean cure AUC, 0.7124, under both latent-count laws. Together, the two empirical applications reinforce the simulation message that no single predictor is uniformly best and that the preferred predictor and kernel depend on the predictive target and data structure.

Several features of the cure-model setting limit how far these results can be interpreted. Cure fractions are only weakly identified from right-censored data with finite follow-up, because a long plateau of censored survivors is consistent with a range of cure probabilities. The melanoma follow-up ends in 1977 and the estimated cure fractions between 0.577 and 0.648 are model-based extrapolations, not quantities read directly from the data. The comparison of split-averaged predicted survival with the Kaplan-Meier curve is a calibration check on the population survival function and does not identify which individuals are cured. The analysis therefore makes no claim to identify individual cure status, and simulation calibration slopes near identity do not change that limitation.

Kernel choice and its hyperparameters shape the estimated intensity because they define which subjects are similar and how rapidly the fitted surface may change. In the primary three-surface comparisons, multiple GG-KM kernel families were evaluated, with their hyperparameters selected from prespecified grids by leakage-free inner tuning; the high-dimensional analysis likewise compared multiple kernel families. Bandwidths that are too small can track idiosyncratic training variation, whereas overly large bandwidths approach a much smoother response; both effects depend on the training-fold scaling used to measure distance. The high-dimensional results further illustrate that kernel family can materially affect performance, with polynomial GG-KM performing particularly well on the interaction surface. A different kernel family or hyperparameter range could therefore shift the balance between GG-KM and the spline, particularly on surfaces whose geometry does not align well with a given kernel, as the periodic surface illustrates. Kernel and penalty selection are model-selection steps, and uncertainty from that selection is not represented by the fitted curve alone. The comparisons fixed a common tuning budget and evaluation protocol for all methods, so the reported differences describe performance at a shared and modest search intensity rather than each method's best attainable configuration.

The kernel-machine estimator also carries a computational cost that grows with sample size. An $n_{\text{train}} \times n_{\text{train}}$ Gram matrix requires quadratic storage, and dense factorization can require

TABLE 5 | Predictive performance in 50 paired repeated holdouts of the breast cancer cohort from Paraíba State, Brazil. Values for the IPCW integrated Brier score (IBS), cure-discrimination area under the curve (AUC_{cure}), and mean predicted cure fraction are means across successful splits, with split standard deviations in parentheses. Convergence is reported as the number of converged fits among successful fits. Runtime is the mean with the maximum in parentheses.

Predictor	Count law	Converged	IPCW IBS	AUC_{cure}	Mean cure prediction	Runtime (s)
Mixture cure	Mixture cure	48/50	0.0883 (0.0019)	0.7851 (0.0156)	0.814 (0.029)	34.04 (132.92)
Promotion time	Poisson	50/50	0.0883 (0.0022)	0.7775 (0.0072)	0.816 (0.009)	38.06 (71.25)
Spline	Poisson	50/50	0.0877 (0.0019)	0.7707 (0.0067)	0.816 (0.009)	1402.30 (2462.32)
Spline	Negative binomial	50/50	0.0877 (0.0019)	0.7707 (0.0067)	0.816 (0.009)	1593.56 (4530.27)
GG-KM (linear)	Poisson	50/50	0.0956 (0.0008)	0.5007 (0.0015)	0.884 (0.000)	137.77 (169.54)
GG-KM (linear)	Negative binomial	50/50	0.0897 (0.0017)	0.7085 (0.0103)	0.818 (0.008)	1934.49 (7967.88)
GG-KM (RBF)	Poisson	50/50	0.0927 (0.0020)	0.6262 (0.0699)	0.807 (0.011)	6648.94 (11951.68)
GG-KM (RBF)	Negative binomial	50/50	0.0927 (0.0020)	0.6262 (0.0699)	0.807 (0.011)	5364.04 (11483.93)
GG-KM (Laplacian)	Poisson	50/50	0.0921 (0.0021)	0.6745 (0.0557)	0.801 (0.012)	5410.98 (9729.17)
GG-KM (Laplacian)	Negative binomial	50/50	0.0921 (0.0021)	0.6745 (0.0557)	0.801 (0.012)	5701.58 (12885.37)
GG-KM (exponential)	Poisson	50/50	0.0901 (0.0022)	0.7124 (0.0579)	0.804 (0.013)	11443.53 (23233.80)
GG-KM (exponential)	Negative binomial	50/50	0.0901 (0.0022)	0.7124 (0.0579)	0.804 (0.013)	8428.04 (17262.88)
GG-KM (Cauchy)	Poisson	50/50	0.0899 (0.0019)	0.6927 (0.0240)	0.798 (0.011)	12069.98 (25675.73)
GG-KM (Cauchy)	Negative binomial	50/50	0.0899 (0.0019)	0.6927 (0.0240)	0.798 (0.011)	8726.52 (19034.95)
GG-KM (polynomial)	Poisson	37/37	0.0927 (0.0013)	0.6721 (0.0375)	0.885 (0.004)	630.28 (717.68)
GG-KM (polynomial)	Negative binomial	36/36	0.0902 (0.0015)	0.7019 (0.0090)	0.820 (0.007)	2973.05 (6276.62)
GG-KM (wavelet)	Poisson	50/50	0.0944 (0.0011)	0.5456 (0.0274)	0.807 (0.012)	5598.89 (10148.10)
GG-KM (wavelet)	Negative binomial	50/50	0.0944 (0.0011)	0.5456 (0.0274)	0.807 (0.012)	5417.58 (13914.95)

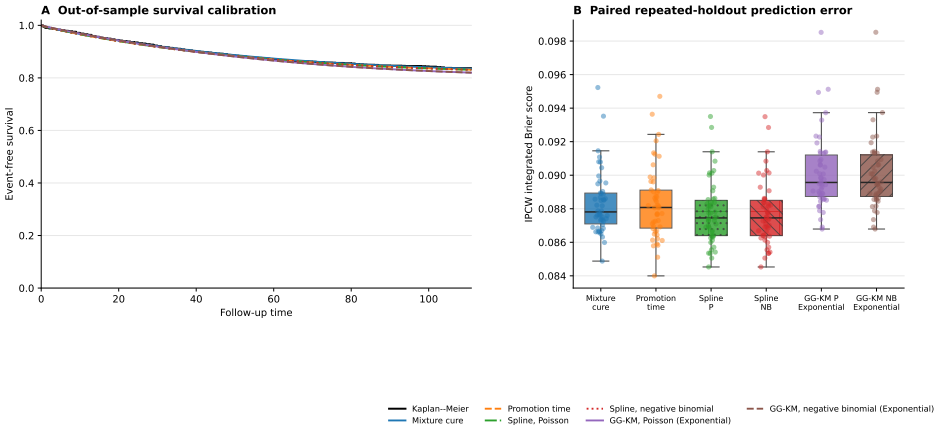


FIGURE 8 | Predictive assessment in 50 paired repeated holdouts of the breast cancer cohort. Panel A compares the full-cohort Kaplan–Meier curve with split-averaged out-of-sample survival predictions over the common supported horizon. Panel B shows split-level IPCW integrated Brier scores and Panel C shows the corresponding cure-discrimination area under the curve (AUC_{cure}). Each point represents one outer split, and lower IBS values indicate better prediction whereas larger cure AUC values indicate better discrimination.

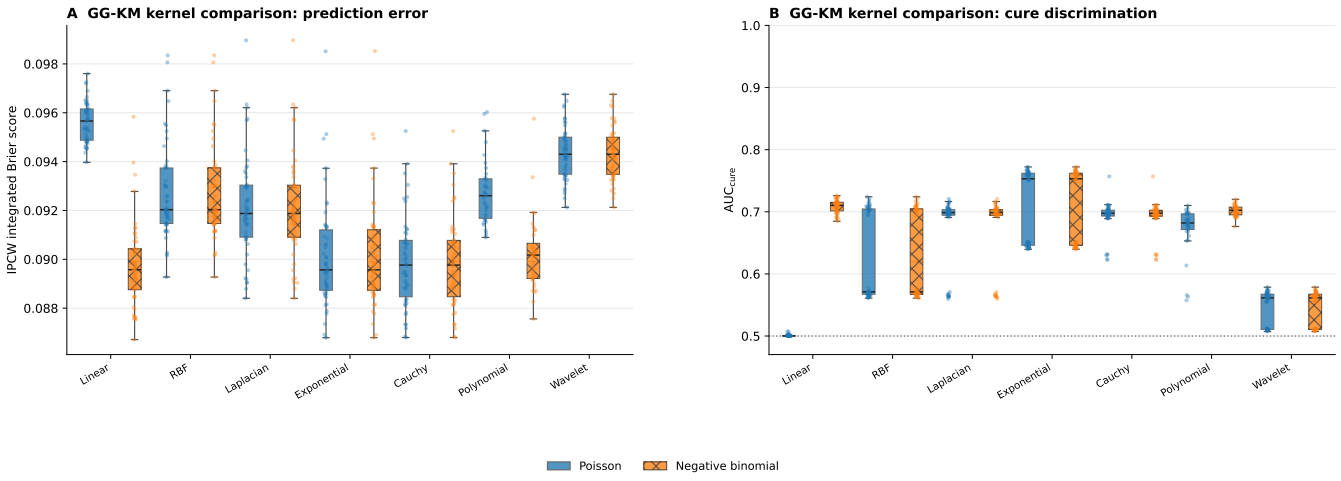


FIGURE 9 | Kernel-specific predictive performance of the GG-KM models in 50 paired repeated holdouts of the breast cancer cohort. Panel A shows the split-level IPCW integrated Brier score and Panel B the cure-discrimination area under the curve (AUC_{cure}) across the seven candidate kernels. Within each kernel, Poisson and negative-binomial GG-KM results are shown side by side, with color and hatching distinguishing the two latent-count laws. Individual points represent outer splits and boxplots summarize their distributions.

cubic time; nested tuning repeats those operations across candidate kernels and inner folds. Reusing a Gram matrix within a candidate and retained eigenspace avoids unnecessary work, but matrices cannot be shared across folds whose preprocessing and training subjects differ. In the primary simulation experiment, these costs remained manageable at the sample sizes studied, with mean per-fit runtimes across methods ranging from 1.13 to 40.32 seconds. The richer breast-cancer application was substantially more computationally demanding, particularly for several nonlinear GG-KM kernels and the spline models, underscoring that computational burden can increase sharply with a larger and more complex predictor representation; these timings do not establish scalability to substantially larger cohorts. Low-rank, Nyström, random-feature, or inducing-point approximations could reduce the burden, although each changes the effective penalty and out-of-sample predictor and would require separate numerical validation. Sample size affected the strength of the conclusions as well as runtime: increasing the sample from 300 to 600 reduced Monte Carlo uncertainty and resolved the quadratic and periodic contrasts more clearly.

Censoring and the event definition further bound the interpretation. The evaluation used inverse-probability-of-censoring weighting with a horizon confined to the region of reliable censoring support, which controls bias from dependent censoring only to the extent that the censoring model is adequate. In the melanoma cohort, deaths from other causes were treated as censored. In the breast-cancer cohort, death from breast cancer was treated as the event, while observations without a recorded breast-cancer death were censored. Accordingly, both empirical analyses are cause specific in their event definition and are not competing-risks analyses. Within these latent cause-specific processes, $p_c(\boldsymbol{x})$ describes the probability of no disease-specific death from the modelled process; it is neither an all-cause survival probability nor a competing-risks cumulative incidence. The interpretation relies on the corresponding independent-censoring assumption. The negative-binomial dispersion was only weakly identified from cure and latency information: under Poisson generation the estimated dispersion correctly stayed near zero, but under true dispersion of 1.00 it was systematically underestimated, ranging from 0.053 to 0.239. This weak identifiability explains the frequent contact of the dispersion estimate with its lower bound and cautions against interpreting the fitted dispersion as a precise measure of overdispersion.

Both empirical applications are observational, and no causal interpretation is attached to the covariate effects or to the predicted cure fractions. The reported associations describe out-of-sample predictive behaviour under a fixed analysis protocol, not the effect of intervening on any covariate.

Future work could extend the estimator in directions that were deliberately left outside the present scope. A controlled comparison against reproducible support-vector or deep cure models would sharpen the positioning of kernel machines among nonlinear approaches, but a fair comparison requires public implementations of the same promotion-time model with common generalized gamma latency, which were not available, so such comparators are noted as open rather than completed. Covariate-dependent latency, low-rank Gram approximations for larger cohorts, adaptive or multiple-kernel methods, broader kernel hyperparameter searches, and joint modelling of the censoring mechanism are natural extensions. The current contribution is the validated combination of an RKHS latent-cause intensity, generalized gamma latency, Poisson and negative-binomial latent counts, penalized EM and ECME estimation, and reproducible leakage-free evaluation with paired Monte Carlo uncertainty.

7 | Declarations

Data and code availability. The malignant melanoma data are distributed as `boot::melanoma` in version 1.3-32 of the `boot` R package, whose official CRAN metadata report an Unlimited license, and are cited to the package documentation and original source. The reproducible article workspace and code used in this study are publicly available at https://github.com/cowvin0/ggkm_paper. The `scikit-cure` Python library, which provides the implementation of the GG-KM model and the cure-modeling framework used in the analyses, is publicly available at https://github.com/cowvin0/scikit-cure.

Ethics and consent. This study reanalyzes a publicly distributed historical cohort and does not report newly collected participant data. The available thesis and package documentation do not establish whether a new institutional determination is required for this secondary analysis. **Author confirmation required:** provide the journal-appropriate ethics approval, exemption, or not-applicable statement and confirm whether a consent statement is required.

Funding, competing interests, and acknowledgments. This study was supported by the Espírito Santo Research and Innovation Support Foundation – FAPES (grant number FAPES nº 1432/2025 P: 2025-R8KXL). Research carried out using the computational resources of the Center for Mathematical Sciences Applied to Industry (CeMEAI) funded by FAPESP (grant 2013/07375-0). The authors declare no competing interests. **Author confirmation required:** supply any additional funding agencies and grant numbers, if applicable, and approve the final acknowledgment and competing-interest statements.

Author contributions. The authorship metadata, namely contributor identities, order, affiliations, and ORCID identifiers, were supplied by the corresponding author, Vera L. D. Tomazella. No contributor has been added, removed, or reordered during preparation of the article workspace.

Use of generative artificial intelligence. Generative artificial-intelligence tools were used during preparation and revision of this article for language editing, code drafting, and consistency checks. Scientific assumptions, analysis choices, citations, executable outputs, and final wording remain subject to author verification, and the confirmed authors retain full responsibility for the submitted work. **Author confirmation required:** confirm the specific tool identities and dates of use, verify this disclosure against current Wiley policy, add any other qualifying tools or uses, and approve the final wording.

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